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One-loop analytic computation of the energy–momentum tensor for lattice gauge theories

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We give a completely analytic determination of the conserved energy–momentum tensor for lattice QCD at one loop. The computation is presented for all covariant gauges, for a general class of one-plaquette discretizations and for a general choice of the discretization of the naive tensor. We review the general strategy for comparing lattice operators with their continuum equivalent ones and present some general results which can be of help for future analytical studies on the lattice. The trace anomaly is verified. As a byproduct we get analytical expressions for the operators appearing in the analysis of deep-inelastic scattering. These expressions had been previously studied only numerically.

1. Introduction

In this paper we carry further the program of defining the energy–momentum tensor (EMT) for a lattice field theory by applying the method of our previous work [1–3]. We address here the case of pure non-abelian gauge theory. This case is of direct physical relevance and moreover, due to asymptotic freedom, the coefficients computed perturbatively are the true corrections in the continuum limit.

The problem for non-abelian gauge theory is more complex than e.g. quantum electrodynamics due to the nonlinearity of the gauge-field equations which causes the appearance, beside gauge invariant operators, of the gluon equations of motion and of BRS-variations [4].

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Even though the structure of the vertices is far more complex than in the abelian case, we shall present here a completely analytic computation. This gives as an advantage the possibility to perform a large number of checks of the computation both at the intermediate steps and on the final results.

In sect. 2 we review the definition of the EMT on the continuum and the Ward identities related to its conservation, which are the key-ingredients of the subsequent analysis.

In sect. 3 we discuss how to define the EMT on the lattice using the Ward identities for its conservation when the lattice spacing is removed.

In sect. 4 we present the general strategy of the analytical computation while in sect. 5 we report a detailed description of our results.

In sect. 6 we discuss an alternative way to derive the energy–momentum tensor on the lattice, translating the continuum operators into their lattice analogues. Exploiting the results of sect. 5 we obtain the same final expression for the EMT.

In appendix A we discuss the ξ-dependence of the renormalization constants, verifying at one-loop that, independently of the lattice lagrangian and of the explicit discretization of the operators, the ξ-dependence is confined only to the coefficients of the operators which vanish on-shell.

In appendix B we discuss the computation of the correlation functions of the gauge-fixing and ghost contribution to the energy–momentum tensor.

In appendix C we discuss systematically the computation of one-loop lattice integrals. We show that every one-loop diagram in pure QCD can be expressed in terms of only three constants of which only two survive in finite results.

2. Ward identities and BRS-invariance

We adopt for the EMT the same definition as given in ref. [2,5] which is obtained by coupling the theory with an external gravitational field [6]. The action is given by

$$S = \int eL \, \mathrm{d}^4 x \tag{2.1}$$

where in our case

$$\begin{aligned} L = {} & \bar{\psi}\gamma^c e^{\lambda}{}_c\left(D_\lambda + \tfrac{1}{2}\sigma^{ab}\omega^{ab}{}_\lambda\right)\psi + m\bar{\psi}\psi + \tfrac{1}{4}F_{\mu\nu}F_{\rho\tau}g^{\mu\rho}g^{\nu\tau} \\ & + (1/2\xi)\left(\nabla_\mu A^\mu\right)^2 - g^{\mu\nu}\bar{c}\nabla_\mu D_\nu c, \end{aligned} \tag{2.2}$$

where

$$D_\mu = \partial_\mu + igA_\mu, \tag{2.3}$$

and

$$D_\mu = \partial_\mu + ig[A_\mu, \quad], \tag{2.4}$$

for the fundamental and adjoint representations respectively; ∇_μ is the covariant derivative in the metric $g_{\mu\nu}$

$$\nabla_\mu A^\mu = \partial_\mu A^\mu + \Gamma_\mu{}^\nu{}_\lambda A^\lambda, \tag{2.5}$$

with $\Gamma_\mu{}^\nu{}_\lambda$ the Christoffel connection; but in particular

$$\nabla_\mu A^\mu = (1/e)\partial_\mu(eA^\mu); \tag{2.6}$$

$\omega^{ab}{}_\lambda$ is the spin connection and

$$\sigma^{ab} = \tfrac{1}{4}[\gamma^a, \gamma^b]. \tag{2.7}$$

$e^c{}_\mu$ is the vierbein related to $g_{\mu\nu}$ by

$$e^a{}_\mu e^a{}_\nu = g_{\mu\nu}, \tag{2.8}$$

and

$$e = \det(e^a{}_\mu). \tag{2.9}$$

Color indices are understood.

Define

$$\Theta^\mu{}_a = -\frac{1}{e}\frac{\delta S}{\delta e^a{}_\mu} \tag{2.10}$$

and the EMT $T^{\mu\nu}$ as the symmetric part of $\Theta^{\mu\nu}$

$$T^{\mu\nu} = \tfrac{1}{2}(\Theta^\mu{}_a e^{a\nu} + \Theta^\nu{}_a e^{a\mu}). \tag{2.11}$$

Even though L is BRS-invariant, i.e. invariant under the transformations

$$\begin{aligned}
\delta A_\mu &= \lambda D_\mu c, \\
\delta c &= -\tfrac{1}{2}ig\lambda[c, c], \\
\delta\bar{c} &= \lambda(1/\xi)\nabla_\rho A^\rho, \\
\delta\psi &= -ig\lambda c\psi, \\
\delta\bar{\psi} &= +ig\bar{\psi}\lambda c,
\end{aligned} \tag{2.12}$$

with λ an anticommuting parameter, $\Theta^{\mu}{}_{a}$ is BRS-invariant only modulo the equation of motion of the ghost. Indeed, let s be the BRS-operator, defined as the left-derivative with respect to the anticommuting parameter of a BRS-variation, then

$$\frac{\delta}{\delta e^{a}{}_{\mu}} sS = s\frac{\delta S}{\delta e^{a}{}_{\mu}} + \frac{\delta(s\bar{c})}{\delta e^{a}{}_{\mu}}\frac{\delta S}{\delta\bar{c}} = 0, \tag{2.13}$$

so that

$$s\Theta^{\mu}{}_{a} = -\frac{1}{e}\frac{\delta(s\bar{c})}{\delta e^{a}{}_{\mu}}\frac{\delta S}{\delta\bar{c}}. \tag{2.14}$$

By collecting the various contributions in S we can write [5]

$$T_{\mu\nu} = T^{\mathrm{F}}_{\mu\nu} + T^{\gamma}_{\mu\nu} + T^{\mathrm{gf}}_{\mu\nu} + T^{\mathrm{gh}}_{\mu\nu}, \tag{2.15}$$

where on the flat background

$$T^{\mathrm{F}}_{\mu\nu} = \tfrac{1}{4}\left(\bar{\psi}\gamma_{\mu}\overleftrightarrow{D}_{\nu}\psi + \bar{\psi}\gamma_{\nu}\overleftrightarrow{D}_{\mu}\psi\right) - \delta_{\mu\nu}\left(\tfrac{1}{2}\bar{\psi}\gamma_{\lambda}\overleftrightarrow{D}_{\lambda}\psi + m\bar{\psi}\psi\right) \tag{2.16}$$

$$= \tilde{T}_{\mu\nu} - \delta_{\mu\nu}L^{\mathrm{F}}, \tag{2.17}$$

with

$$\overleftrightarrow{D}_{\mu} = \partial_{\mu} + igA_{\mu} - \overleftarrow{\partial}_{\mu} + igA_{\mu}, \tag{2.18}$$

and

$$T^{\gamma}_{\mu\nu} = F_{\mu\alpha}F_{\nu\alpha} - \tfrac{1}{4}\delta_{\mu\nu}F_{\alpha\beta}F_{\alpha\beta}, \tag{2.19}$$

$$T^{\mathrm{gf}}_{\mu\nu} = (1/\xi)\Big[-A_{\mu}\partial_{\nu}\partial_{\lambda}A_{\lambda} - A_{\nu}\partial_{\mu}\partial_{\lambda}A_{\lambda} + \delta_{\mu\nu}\left(\tfrac{1}{2}(\partial_{\lambda}A_{\lambda})^{2} + A_{\lambda}\partial_{\lambda}\partial_{\rho}A_{\rho}\right)\Big], \tag{2.20}$$

$$T^{\mathrm{gh}}_{\mu\nu} = \partial_{\mu}\bar{c}D_{\nu}c + \partial_{\nu}\bar{c}D_{\mu}c - \delta_{\mu\nu}\partial_{\lambda}\bar{c}D_{\lambda}c. \tag{2.21}$$

$T^{\mathrm{F}}_{\mu\nu}$ and $T^{\gamma}_{\mu\nu}$ are gauge invariant, and therefore invariant also under the BRS transformations. The sum of the two non-gauge-invariant terms $T^{\mathrm{gf}}_{\mu\nu} + T^{\mathrm{gh}}_{\mu\nu}$ can be written as the BRS variation of the operator $\Xi_{\mu\nu}$, given explicitly by

$$\Xi_{\mu\nu} = -A_{\mu}\partial_{\nu}\bar{c} - A_{\nu}\partial_{\mu}\bar{c} + \delta_{\mu\nu}\left[\tfrac{1}{2}\partial_{\lambda}A_{\lambda}\bar{c} + A_{\lambda}\partial_{\lambda}\bar{c}\right] \tag{2.22}$$

plus a term proportional to the equation of motion of the ghost. With these definitions we have for the gluonic part that

$$\int \mathrm{d}^4x \; T_{\mu\mu} = 0. \tag{2.23}$$

The canonical energy–momentum tensor, simply defined as the generator of translations, is neither symmetric nor gauge invariant, due to the presence of non-scalar fields.

The Ward identities of the 1-particle irreducible Green functions are

$$\begin{aligned}\partial_\mu T_{\mu\nu}(A, \bar{\psi}, \psi, \bar{c}, c) = &-\frac{\delta\Gamma}{\delta c}\partial_\nu c - \partial_\nu \bar{c}\frac{\delta\Gamma}{\delta\bar{c}} - \partial_\nu A_\mu \frac{\delta\Gamma}{\delta A_\mu} + \partial_\mu\left(A_\nu \frac{\delta\Gamma}{\delta A_\mu}\right)\\ &-\frac{\delta\Gamma}{\delta\psi}\partial_\nu\psi - \partial_\nu\bar{\psi}\frac{\delta\Gamma}{\delta\bar{\psi}}\\ &+\tfrac{1}{2}\partial_\mu\left(\frac{\delta\Gamma}{\delta\psi}\sigma^{\mu\nu}\psi - \bar{\psi}\sigma^{\mu\nu}\frac{\delta\Gamma}{\delta\bar{\psi}}\right).\end{aligned} \tag{2.24}$$

We stress the linear dependence of the divergence $\partial_\mu T_{\mu\nu}$ on the elementary fields and on the vertex functions. This assures that the divergencies of $T_{\mu\nu}$ are directly related to the divergencies of the elementary Green functions, which is crucial for the renormalization of the energy–momentum tensor [7,8].

In this paper we shall concentrate only on the gluonic sector as the fermion sector of QCD at one loop is identical, apart from some trivial Casimir factor, to the QED case, which has been dealt with in detail in ref. [2].

For the case of the Green function with two external gluons (2.24) reads in momentum space

$$\begin{aligned}&\sum_\mu (p_\mu + q_\mu)\langle T_{\mu\nu}(-p-q)A_\alpha(p)A_\beta(q)\rangle^{\text{irr.}}\\ &= \text{tree} + \sum_\mu \left\{q_\mu\left[\delta_{\mu\nu}\Pi_{\alpha\beta}(p) - \delta_{\beta\nu}\Pi_{\alpha\mu}(p)\right] + p_\mu\left[\delta_{\mu\nu}\Pi_{\alpha\beta}(q) - \delta_{\alpha\nu}\Pi_{\mu\beta}(q)\right]\right\},\end{aligned} \tag{2.25}$$

where $\Pi_{\alpha\beta}(p)$ is the gluon self-energy. Because of BRS-invariance we have also that

$$\sum_{\alpha,\beta} p^\alpha q^\beta \langle T_{\mu\nu}(-p-q)A_\alpha(p)A_\beta(q)\rangle^{\text{irr.}} = 0. \tag{2.26}$$

Indeed, let us take the identity

$$
\begin{aligned}
0 &= s\big\langle T_{\mu\nu}(-p-q)\partial^{\alpha}A_{\alpha}(p)\bar{c}(q)\big\rangle \\
&= \big\langle\big(sT_{\mu\nu}(-p-q)\big)\partial^{\alpha}A_{\alpha}(p)\bar{c}(q)\big\rangle + \big\langle T_{\mu\nu}(-p-q)\partial^{\alpha}D_{\alpha}c(p)\bar{c}(q)\big\rangle \\
&\quad + \big\langle T_{\mu\nu}(-p-q)\partial^{\alpha}A_{\alpha}(p)(1/\xi)\partial^{\rho}A_{\rho}(q)\big\rangle.
\end{aligned}
\tag{2.27}
$$

The first two terms are proportional to the equation of motion of the ghost and as such give rise only to disconnected contributions. Therefore going over to the connected contributions we get (2.26).

By taking the derivative of (2.25) with respect to the momentum transfer $p_{\mu} + q_{\mu}$, and going to zero momentum transfer ($q = -p$), we get

$$
\begin{aligned}
\langle T_{\mu\nu}(0)A_{\alpha}(p)A_{\beta}(-p)\rangle^{\text{irr.}} &= \langle T_{\mu\nu}(0)A_{\alpha}(p)A_{\beta}(-p)\rangle^{\text{tree}} \\
&\quad - \tfrac{1}{2}\big[\delta_{\mu\beta}\Pi_{\nu\alpha} + \delta_{\mu\alpha}\Pi_{\nu\beta} + \delta_{\nu\alpha}\Pi_{\mu\beta} + \delta_{\nu\beta}\Pi_{\mu\alpha}\big] \\
&\quad + \delta_{\mu\nu}\Pi_{\alpha\beta} - \tfrac{1}{2}\left(p_{\mu}\frac{\partial}{\partial p_{\nu}} + p_{\nu}\frac{\partial}{\partial p_{\mu}}\right)\Pi_{\alpha\beta}.
\end{aligned}
\tag{2.28}
$$

For the case of an external pair of ghost–antighost at zero momentum transfer we get

$$
\langle T_{\mu\nu}(0)c(-p)\bar{c}(p)\rangle^{\text{irr.}} = -\delta_{\mu\nu}\Delta^{-1}(p) + \tfrac{1}{2}\left(p_{\mu}\frac{\partial}{\partial p_{\nu}} + p_{\nu}\frac{\partial}{\partial p_{\mu}}\right)\Delta^{-1}(p), \tag{2.29}
$$

where $\Delta(p)$ is the ghost propagator.

3. Definition of the energy–momentum tensor on the lattice

Let us now pass to discuss the lattice theory: we will study an SU(N) gauge theory with the lagrangian [9]

$$
L^{\gamma} = -\frac{1}{2a^4 g^2}\sum_{\mu \neq \nu} \mathrm{Tr}\, P(x;\mu,\nu), \tag{3.1}
$$

with

$$
P(x;\mu,\nu) = U(x, x+\hat{\mu})U(x+\hat{\mu}, x+\hat{\mu}+\hat{\nu})U(x+\hat{\mu}+\hat{\nu}, x+\hat{\nu})U(x+\hat{\nu}, x), \tag{3.2}
$$

and

$$U(x, x+\hat{\mu}) = \exp\left(iagT^c A^c_\mu(x)\right), \tag{3.3}$$

with

$$\mathrm{Tr}(T^a T^b) = \tfrac{1}{2}\delta^{ab}. \tag{3.4}$$

For a perturbative calculation one has to add to the lagrangian a gauge-fixing and a ghost-term. We choose [10]

$$L^{\mathrm{gf}} = \frac{1}{2a^2\xi}\mathrm{Tr}\left[\sum_\mu A_\mu(x) - A_\mu(x-\hat{\mu})\right]^2, \tag{3.5}$$

while

$$L^{\mathrm{gh}} = \frac{1}{a^2}\sum_\mu \left(\bar{c}^a(x+\hat{\mu}) - \bar{c}^a(x)\right)$$
$$\times\left\{E^{-1}_{ba}\left(agA_\mu(x)\right)c^b(x+\hat{\mu}) - E^{-1}_{ab}\left(agA_\mu(x)\right)c^b(x)\right\}. \tag{3.6}$$

The lagrangian is invariant under the BRS transformations, which on the lattice take the form (2.12) on flat background except for the following two variations

$$\delta\bar{c}(x) = (\lambda/\xi)\sum_\mu\left[A_\mu(x) - A_\mu(x-\hat{\mu})\right], \tag{3.7}$$

$$\delta A^a_\mu(x) = (\lambda/g)E^{-1}_{ba}\left(agA_\mu(x)\right)c^b(x+\hat{\mu}) - (\lambda/g)E^{-1}_{ab}\left(agA_\mu(x)\right)c^b(x), \tag{3.8}$$

where

$$E_{ab}(\phi) = \left[\frac{\mathrm{e}^{i\tilde{\phi}} - 1}{i\tilde{\phi}}\right]_{ab} \tag{3.9}$$

with

$$i\tilde{\phi}_{ab} = \sum_c f_{abc}\phi_c. \tag{3.10}$$

For the Feynman rules we refer to ref. [10] *.

* However we do not agree with the four-gluon vertex (A.7) given in their appendix A, because of the two terms in which the restriction $\Sigma_{\sigma\neq\mu}$ appears. We find that the terms with $\sigma = \mu$ must be there as it can be checked from the Ward identity which relates the three-gluon to the four-gluon vertex. See appendix A, eq. (A.2).

One could consider more general one-plaquette actions; at one loop the relevant perturbative lagrangian is parametrized by two constants c_1 and c_2 defined by the formula [11]

$$L^{\gamma} = -\tfrac{1}{2}\sum_{\mu\nu}\mathrm{Tr}\; G_{\mu\nu}^2 + \tfrac{1}{24}g^2 c_1 \sum_{\mu\nu}\mathrm{Tr}\; G_{\mu\nu}^4 + g^2 c_2 \sum_{\mu\nu}\left(\mathrm{Tr}\; G_{\mu\nu}^2\right)^2 + \mathrm{O}(g^4), \quad (3.11)$$

where

$$P(x;\mu,\nu) = \exp\left[igG_{\mu\nu}(x)\right]. \quad (3.12)$$

For Wilson's action [9] $c_1 = 1$ and $c_2 = 0$, for Manton's [12] and the heat-kernel action [13] $c_1 = c_2 = 0$, and for the adjoint-representation action [14] $c_1 = 1$ and $c_2 = 1/8$. Let us notice however that the results of our computation will depend only on a combination of c_1 and c_2 and we will express our formulas in terms of

$$c = c_1 + \frac{24N(N^2+1)}{2N^2-3}c_2. \quad (3.13)$$

Let us now define the energy–momentum tensor. According to our procedure we are free to choose the lattice discretization of the EMT. We define

$$\begin{aligned} F_{\mu\nu}(x) = \frac{1}{8agi}\{[&P(x;\mu,\nu) + P(x;\nu,-\mu) + P(x;-\mu,-\nu) \\ &+P(x;-\nu,\mu) - P(x;\nu,\mu) - P(x;\mu,-\nu) \\ &-P(x;-\nu,-\mu) - P(x;-\mu,\nu)]\} \end{aligned} \quad (3.14)$$

and

$$F_{\mu\nu}^a(x) = 2\;\mathrm{Tr}\left[T^a F_{\mu\nu}(x)\right]. \quad (3.15)$$

Our choice [2] for the lattice version of the gauge-invariant part of $T_{\mu\nu}^{\mathrm{tree}}$ has the same form as in the continuum with

$$\sum_a F_{\mu\alpha}^a F_{\nu\alpha}^a = 4\sum_a \mathrm{Tr}\left(T^a F_{\mu\alpha}\right)\,\mathrm{Tr}\left(T^a F_{\nu\alpha}\right). \quad (3.16)$$

We remark that this definition differs at higher-loop but not at one-loop level from

$$\sum_a F_{\mu\alpha}^a F_{\nu\alpha}^a = 2\;\mathrm{Tr}\left(F_{\mu\alpha}F_{\nu\alpha}\right), \quad (3.17)$$

which is used elsewhere [15,16]. Indeed, the T^c form a complete basis for the hermitian traceless matrices of dimension N, but $F_{\mu\nu}$ is not necessarily traceless, so that

$$F_{\nu\alpha} = 2T^a \operatorname{Tr}(T^a F_{\nu\alpha}) + (1/N)\mathbb{1} \operatorname{Tr}(F_{\nu\alpha}), \tag{3.18}$$

and thus

$$2 \operatorname{Tr}(F_{\mu\alpha} F_{\nu\alpha}) = 4 \sum_a \operatorname{Tr}(T^a F_{\mu\alpha}) \operatorname{Tr}(T^a F_{\nu\alpha}) + (2/N) \operatorname{Tr}(F_{\mu\alpha}) \operatorname{Tr}(F_{\nu\alpha}). \tag{3.19}$$

But $F_{\alpha\mu}$ is antisymmetric in α, μ and thus the first contribution to Tr $F_{\alpha\mu}$ is of order g^3 and therefore the second term will show up only at the two-loop level.

More general choices of $T_{\mu\nu}^{\text{tree}}$ can be obtained by replacing into the definition of $F_{\mu\nu}$, formula (3.14), $P(x; \mu, \nu)$ with $f(P(x; \mu, \nu) - 1)$, where f is an arbitrary function with $f'(0) = 1$. At one loop the relevant part is given by

$$\begin{aligned} f(P(x; \mu, \nu) - 1) &= P(x; \mu, \nu) - 1 + \alpha_1 (P(x; \mu, \nu) - 1)^2 \\ &\quad + \alpha_2 (P(x; \mu, \nu) - 1)^3 + \dots, \end{aligned} \tag{3.20}$$

where α_1 and α_2 are arbitrary constants.

The gauge-fixing and ghost contributions to $T_{\mu\nu}^{\text{tree}}$ have been taken as the BRS-variation of a lattice form of $\Xi_{\mu\nu}$ which obeys exactly the condition to be traceless at zero momentum [2] like in the continuum

$$\sum_x \sum_\mu \Xi_{\mu\mu}(x) = 0. \tag{3.21}$$

We have

$$\begin{aligned} \Xi_{\mu\nu}(x) = {} & \frac{A_\mu(x) + A_\mu(x - \hat{\mu})}{2} \bar{\Delta}_\nu \bar{c}(x) + \frac{A_\nu(x) + A_\nu(x - \hat{\nu})}{2} \bar{\Delta}_\mu \bar{c}(x) \\ & - \delta_{\mu\nu} \sum_\lambda \left[\tfrac{1}{2} \bar{\bar{\Delta}}_\lambda A_\lambda(x) \bar{c}(x) + \frac{A_\lambda(x) + A_\lambda(x - \hat{\lambda})}{2} \bar{\Delta}_\lambda \bar{c}(x) \right], \end{aligned} \tag{3.22}$$

where

$$\bar{\Delta}_\lambda \phi(x) = (1/2a)\left[\phi(x + \hat{\lambda}) - \phi(x - \hat{\lambda})\right], \tag{3.23}$$

and

$$\begin{aligned} \bar{\bar{\Delta}}_\lambda \phi(x) &= (1/4a)\left[\phi(x + \hat{\lambda}) - \phi(x - \hat{\lambda}) + \phi(x) - \phi(x - 2\hat{\lambda})\right] \\ &= \tfrac{1}{2} \bar{\Delta}_\lambda \left[\phi(x) + \phi(x - \hat{\lambda})\right]. \end{aligned} \tag{3.24}$$

Our choice of $T_{\mu\nu}^{\text{tree}}$ as any other local discretization of the EMT of the continuum is not conserved. Indeed,

$$\partial_\mu T_{\mu\nu}^{\text{tree}} = R_\nu + X_\nu, \tag{3.25}$$

where R_ν is proportional to the lattice equations of motion and is the lattice transcription of the corresponding terms appearing in the continuum; X_ν is an operator which formally vanishes when $a \to 0$ but due to radiative corrections gives rise to finite contributions when inserted into Green functions. X_ν vanishes at zero external momentum and thus can be re-written as

$$X_\nu = \partial_\mu O_{\mu\nu}, \tag{3.26}$$

with [4] $O_{\mu\nu}$ the sum of generic symmetric and BRS-invariant operators and operators proportional to the equations of motions of dimension less or equal to four *

$$O_{\mu\nu} = \sum_i C_i(g, \xi) O_{\mu\nu}^{[i]}, \tag{3.27}$$

where g is the coupling constant and ξ the gauge-parameter.

Thus we shall define the EMT on the lattice as

$$T_{\mu\nu} = T_{\mu\nu}^{\text{tree}} - O_{\mu\nu}. \tag{3.28}$$

The operators which may contribute to $O_{\mu\nu}$ are

$$\begin{aligned} O_{\mu\nu}^{[1]} &= T_{\mu\nu}^{\text{gi,tree}}, \\ O_{\mu\nu}^{[2]} &= \delta_{\mu\nu}\mathscr{L}^{\text{gi}} = \tfrac{1}{4}\delta_{\mu\nu}\sum_{\alpha,\beta} F_{\alpha\beta}F_{\alpha\beta}, \\ O_{\mu\nu}^{[3]} &= \delta_{\mu\nu}O_{\mu\nu}^{[1]}, \end{aligned} \tag{3.29}$$

which are gauge-invariant; the BRS-variations

$$\begin{aligned} O_{\mu\nu}^{[4]} &= (1/\xi)\left[(\partial_\nu A_\mu + \partial_\mu A_\nu)(\partial_\rho A_\rho) - \tfrac{1}{2}\delta_{\mu\nu}(\partial_\rho A_\rho)^2\right] \\ &\quad - \left(\bar{c}\partial_\mu D_\nu c + \bar{c}\partial_\nu D_\mu c - \tfrac{1}{2}\delta_{\mu\nu}\bar{c}\partial_\rho D_\rho c\right), \\ O_{\mu\nu}^{[5]} &= \delta_{\mu\nu}(\mathscr{L}^{\text{gf}} + \mathscr{L}^{\text{gh}}) = \delta_{\mu\nu}\left[(1/2\xi)(\partial_\rho A_\rho)^2 - \bar{c}\partial_\rho D_\rho c\right], \\ O_{\mu\nu}^{[6]} &= \delta_{\mu\nu}O_{\mu\nu}^{[4]}, \end{aligned} \tag{3.30}$$

* We assume here the renormalizability of composite operators on the lattice, a fact which has not yet been rigorously proven. However there have been recently important advances in this field: power-counting theorems have been proved for the lattice theory and a complete proof of the renormalizability of the Green functions has been given [17].

and

$$O^{[7]}_{\mu\nu} = (1/\xi)\left[-(A_\mu\partial_\nu + A_\nu\partial_\mu)(\partial_\rho A_\rho) + \tfrac{1}{2}\delta_{\mu\nu}A_\rho\partial_\rho\partial_\sigma A_\sigma\right]$$
$$+\left(\partial_\mu\bar{c}D_\nu c + \partial_\nu\bar{c}D_\mu c - \tfrac{1}{2}\delta_{\mu\nu}\partial_\rho\bar{c}D_\rho c\right),$$
$$O^{[8]}_{\mu\nu} = -(1/2\xi)A_\rho\partial_\rho\partial_\sigma A_\sigma + \tfrac{1}{2}\delta_{\mu\nu}\partial_\rho\bar{c}D_\rho c,$$
$$O^{[9]}_{\mu\nu} = \delta_{\mu\nu}O^{[7]}_{\mu\nu}, \tag{3.31}$$

and the terms proportional to the equations of motions (which also include terms which are BRS-variations)

$$O^{[10]}_{\mu\nu} = A_\nu\frac{\delta S}{\delta A_\mu} + A_\mu\frac{\delta S}{\delta A_\nu} - \tfrac{1}{2}\delta_{\mu\nu}\sum_\sigma A_\sigma\frac{\delta S}{\delta A_\sigma},$$
$$O^{[11]}_{\mu\nu} = \tfrac{1}{2}\delta_{\mu\nu}\sum_\sigma A_\sigma\frac{\delta S}{\delta A_\sigma},$$
$$O^{[12]}_{\mu\nu} = \delta_{\mu\nu}O^{[10]}_{\mu\nu}. \tag{3.32}$$

The BRS-invariance of the EMT restricts the possible combinations of these operators. The identity (2.26) imposes

$$C_4 = 0, \qquad C_5 = 0, \qquad C_6 = 0,$$
$$C_7 = -C_{10}, \qquad C_8 = -C_{11}, \qquad C_9 = -C_{12}. \tag{3.33}$$

They mean that there are cancellations of terms which are BRS-variations in the equations of motions and the rest.

The presence of equations of motions is at variance with the case of QED, where it is possible to show that, to all orders in perturbation theory, they cannot appear [2,4,7]. Indeed, as the BRS-transformations are linear for QED, a BRS-invariant operator cannot mix with operators which are not BRS-invariant, like the equations of motions. Thus in QED the only non-zero coefficients can be C_1, C_2 and C_3.

The operators $O^{[3]}$, $O^{[6]}$, $O^{[9]}$ and $O^{[12]}$ do not respect Lorentz covariance but they satisfy the hypercubic symmetry of the lattice. In principle only C_1, C_2 and C_3 are ξ-independent. Contributions to the trace of the EMT can come out from C_2, C_5, C_8 and C_{11}.

The operator $O^{[2]}$ differs from $O^{[8]}$ by a total derivative plus the equation of motions of the ghosts and they cannot be distinguished at zero momentum transfer. Thus to determine all the coefficients of the gauge-invariant operators we have exploited the Ward identities for two-gluon 1-particle irreducible Green functions when the divergence of the EMT is inserted at non-zero momentum. An alternative method would be to work with an insertion at zero momentum but by taking into account also Green functions in the ghost sector [8].

4. Analytic evaluation of the lattice integrals

The Feynman integrals on the lattice will be computed using the method by Kawai et al. [10]. The computation of a quadratically divergent integral of the form

$$I = \int \mathrm{d}k\ \mathscr{I}(k, p, q, a, d), \tag{4.1}$$

where p and q are external momenta, is based on the splitting

$$I = J + (I - J) \tag{4.2}$$

where

$$J = \int \mathrm{d}k\ \mathscr{I}(k, 0, 0, a, d) + \sum_{\rho,\sigma} \left[p_\rho q_\sigma \int \mathrm{d}k \frac{\partial^2 \mathscr{I}(k, p, q, a, d)}{\partial p_\rho \partial q_\sigma} \bigg|_{p=q=0} \right.$$
$$\left. + \frac{p_\rho p_\sigma}{2} \int \mathrm{d}k \frac{\partial^2 \mathscr{I}(k, p, 0, a, d)}{\partial p_\rho \partial p_\sigma} \bigg|_{p=0} + \frac{q_\rho q_\sigma}{2} \int \mathrm{d}k \frac{\partial^2 \mathscr{I}(k, 0, q, a, d)}{\partial q_\rho \partial q_\sigma} \bigg|_{q=0} \right], \tag{4.3}$$

is the Taylor expansion of the graph up to second order in the external momenta. For finite lattice spacing a and p, $q \neq 0$ the integral I is well defined both in the ultraviolet and in the infrared region. However J and $I - J$ are separately infrared divergent and thus in order to consider the two terms independently we must introduce an intermediate infrared regularization: we will use dimensional regularization, assuming $d > 4$, but other infrared regularizations, for instance introducing a mass m in the propagators, would work equally well. In the dimensional scheme the infrared singularities will show up as poles in $\epsilon = 4 - d$ in both J and $I - J$. However these contributions must be such to cancel when J and $I - J$ are added, as no infrared singularity was present in the original integral. This splitting of the original integral is particularly convenient. Indeed, because of the subtraction, $I - J$ is well behaved in the ultraviolet and thus we can take the limit $a \to 0$ and evaluate it in the continuum: notice that if we use dimensional regularization J vanishes in the continuum and thus we are left with $\mathscr{I}(k, p, q, 0, d)$. Also the computation of J on the lattice is easy, as the integral contains only one scale, the lattice spacing.

The computation of J is performed by considering all the possible values of the indices α, β, μ and ν. Once a particular assignment of the indices is given each integral is reduced to the algebra of elementary integrals of the form

$$\mathscr{B}(r; n_x, n_y, n_z, n_t) = \int_{-\pi}^{\pi} \left(\frac{\mathrm{d}k}{2\pi}\right)^d \frac{\hat{k}_x^{2n_x} \hat{k}_y^{2n_y} \hat{k}_z^{2n_z} \hat{k}_t^{2n_t}}{(\hat{k}^2)^r} \tag{4.4}$$

with

$$\hat{k}_x = 2\sin(k_x/2) \tag{4.5}$$

for $n_i = 0, 1, \ldots;\ r = 0, \ldots, 5;\ \Sigma_i n_i \leqslant 7$ and x, y, z, t all different. The computation of such integrals is given in appendix C.

5. Analysis of the graphs

In this section we shall report our analytical results concerning the implementation of our project for the definition of the EMT on the lattice at first order in perturbation theory.

In order to determine the correction terms to the $T_{\mu\nu}^{\text{tree}}$ we shall impose the Ward identities (2.25) for the insertion of the EMT into the Green function with two gluons. Since the continuum theory in dimensional regularization already satisfies the Ward identities, we only need to study the J-part of the various correlation functions. We will begin by computing the J-part (see (4.3)) of the expectation value

$$\left\langle T_{\mu\nu}^{\text{tree}}(-p-q) A_\alpha^a(p) A_\beta^b(q) \right\rangle^{\text{irr.}} \tag{5.1}$$

that is the difference of the value of (5.1) on the lattice and on the continuum in dimensional regularization with $d = 4 - \epsilon$.

Each graph will contribute a term of order zero and one of order two in the external momenta p and q.

At zeroth order in the external momenta the form factors symmetric under the exchanges $\alpha \leftrightarrow \beta$, $\mu \leftrightarrow \nu$ which can appear on the hypercubic lattice are

$$\begin{array}{ll} A_1 & \delta_{\mu\nu}\delta_{\alpha\beta}, \\ A_2 & \delta_{\alpha\mu}\delta_{\beta\nu} + \delta_{\alpha\nu}\delta_{\beta\mu}, \\ B_1 & \delta_{\mu\nu\alpha\beta}, \end{array}$$

where the A's are Lorentz invariant while B_1 is not. Because of gauge invariance, the sum of all the zeroth-order terms must vanish and we have used this property to check, whenever it has been possible, the overall combinatorial factors.

At second order in the external momenta p and q the possible form factors symmetric under the exchanges $(\alpha, p) \leftrightarrow (\beta, q)$, $\mu \leftrightarrow \nu$ are

$$
\begin{array}{ll}
L_1 & (p_\alpha p_\beta + q_\alpha q_\beta)\delta_{\mu\nu}, \\
L_2 & (p^2 + q^2)\delta_{\alpha\beta}\delta_{\mu\nu}, \\
L_3 & (p_\mu p_\nu + q_\mu q_\nu)\delta_{\alpha\beta}, \\
L_4 & p_\alpha p_\mu \delta_{\beta\nu} + q_\beta q_\mu \delta\alpha_\nu + (\text{symm. } \mu \leftrightarrow \nu), \\
L_5 & p_\beta p_\mu \delta_{\alpha\nu} + q_\alpha q_\mu \delta_{\beta\nu} + (\text{symm. } \mu \leftrightarrow \nu), \\
L_6 & (p^2 + q^2)(\delta_{\alpha\mu}\delta_{\beta\nu} + \delta_{\alpha\nu}\delta_{\beta\mu}), \\
L_7 & p_\alpha q_\beta \delta_{\mu\nu}, \\
L_8 & p_\beta q_\alpha \delta_{\mu\nu}, \\
L_9 & p \cdot q \delta_{\alpha\beta}\delta_{\mu\nu}, \\
L_{10} & (p_\mu q_\nu + p_\nu q_\mu)\delta_{\alpha\beta}, \\
L_{11} & p_\alpha q_\mu \delta_{\beta\nu} + q_\beta p_\mu \delta_{\alpha\nu} + (\text{symm. } \mu \leftrightarrow \nu), \\
L_{12} & p_\beta q_\mu \delta_{\alpha\nu} + q_\alpha p_\mu \delta_{\beta\nu} + (\text{symm. } \mu \leftrightarrow \nu), \\
L_{13} & p \cdot q(\delta_{\alpha\mu}\delta_{\beta\nu} + \delta_{\alpha\nu}\delta_{\beta\mu}), \\
I_1 & (p_\alpha^2 + q_\alpha^2)\delta_{\alpha\beta}\delta_{\mu\nu}, \\
I_2 & (p_\mu^2 + q_\nu^2)\delta_{\alpha\beta}\delta_{\mu\nu}, \\
I_3 & (p^2 + q^2)\delta_{\mu\nu\alpha\beta}, \\
I_4 & (p_\alpha^2 + q_\alpha^2)\delta_{\mu\nu\alpha\beta}, \\
I_5 & (p_\alpha^2 + q_\beta^2)(\delta_{\mu\alpha}\delta_{\nu\beta} + \delta_{\mu\beta}\delta_{\nu\alpha}), \\
I_6 & (p_\beta^2 + q_\alpha^2)(\delta_{\nu\alpha}\delta_{\nu\beta} + \delta_{\mu\beta}\delta_{\nu\alpha}), \\
I_7 & p_\alpha p_\beta \delta_{\mu\nu\alpha} + q_\alpha q_\beta \delta_{\mu\nu\beta}, \\
I_8 & p_\alpha p_\beta \delta_{\mu\nu\beta} + q_\alpha q_\beta \delta_{\mu\nu\alpha}, \\
I_9 & (p_\mu p_\nu + q_\nu q_\mu)(\delta_{\mu\alpha\beta} + \delta_{\nu\alpha\beta}), \\
I_{10} & p_\alpha q_\alpha \delta_{\alpha\beta}\delta_{\mu\nu}, \\
I_{11} & p_\mu q_\mu \delta_{\alpha\beta}\delta_{\mu\nu}, \\
I_{12} & p \cdot q \delta_{\alpha\beta\mu\nu}, \\
I_{13} & p_\alpha q_\alpha \delta_{\alpha\beta\mu\nu}, \\
I_{14} & (p_\alpha q_\beta + p_\beta q_\alpha)(\delta_{\mu\alpha}\delta_{\nu\beta} + \delta_{\mu\beta}\delta_{\nu\alpha}), \\
I_{15} & p_\alpha q_\beta(\delta_{\mu\nu\alpha} + \delta_{\mu\nu\beta}), \\
I_{16} & p_\beta q_\alpha(\delta_{\mu\nu\alpha} + \delta_{\mu\nu\beta}), \\
I_{17} & (p_\mu q_\nu + p_\nu q_\mu)(\delta_{\mu\alpha\beta} + \delta_{\nu\alpha\beta}),
\end{array}
$$

where the L's are compatible with Lorentz invariance, while the I's respect only the hypercubic symmetry.

Let us first compute the correlation of $T^{\mathrm{gi}}_{\mu\nu}$. We will begin by considering the terms which are proportional to the gauge parameter ξ. We have found that all these contributions can be computed without making use of the explicit lattice form for the lagrangian and for the tensor. More generally, in appendix A we will show that for the insertion of a gauge-invariant dimension-four operator $\mathscr{O}$

$$\left\langle \mathscr{O}(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\mathrm{irr.}}$$

(i) the ξ^3-term vanish; (ii) the ξ^2-term takes exactly the same value as in the continuum with no J-contribution; (iii) the J-part of the ξ-term is given by (independently from the discretization of both the lagrangian and the operator $\mathscr{O}$) by

$$N\delta^{ab}g^2\frac{\delta^2\mathscr{O}(p+q)}{\delta A^c_\alpha(p)\delta A^c_\beta(q)}\bigg|_{A=0}\left(\frac{X}{32}-\frac{Z_0}{48}\right) \tag{5.2}$$

where *

$$X=-\frac{2}{\pi^2\epsilon}+\frac{1}{\pi^2}(F_0-\log 4\pi). \tag{5.3}$$

We are now left to compute the insertion of $T^{\mathrm{gi,tree}}_{\mu\nu}$ in the gauge $\xi=1$. The relevant graphs are shown in fig. 1. We have reported in tables 1–6 the results for the insertion of $\Sigma_\sigma F_{\mu\sigma}F_{\nu\sigma}$. Rows or columns with all the entries zero have been omitted. Tables 1–4 give contributions which should be multiplied by

$$2\delta^{cd}g^2\ \mathrm{Tr}[T^a,T^c][T^d,T^b]=N\delta^{ab}g^2. \tag{5.4}$$

Tables 5 and 6 give contributions which should be multiplied respectively by

$$\frac{2N^2-3}{24N}c\delta^{ab}g^2, \tag{5.5}$$

and

$$\frac{2N^2-3}{24N}d\delta^{ab}g^2, \tag{5.6}$$

* If we had used instead a mass m in the propagator as an infrared regulator the same results would hold with $X=-(\log m^2-F_0+\gamma_E)/\pi^2$

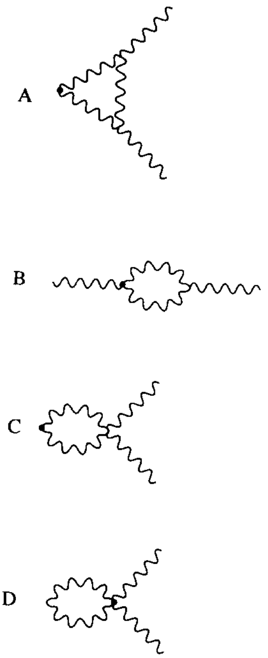

Fig. 1. Feynman graphs for the insertion of $T^{\mathrm{gi}}_{\mu\nu}$ in the two-gluon correlation function.

where

$$\frac{1}{4!}\delta^{cd}\,\mathrm{Tr}\big[\{T^a, T^b\}\{T^c, T^d\} + \{T^a, T^c\}\{T^b, T^d\} + \{T^a, T^d\}\{T^b, T^c\}\big] = \delta^{ab}\frac{2N^2-3}{24N}, \tag{5.7}$$

and

$$d = 1 + 6\alpha_1 + 6\alpha_2, \tag{5.8}$$

where α_1 and α_2 have been defined in eq. (3.20). For the most simple definition, eq. (3.16), and also for the definition used in refs. [15,16], $d = 1$.

Only the graphs (A), (B) and (C) contain $1/\epsilon$ contributions. As we know from the discussion in sect. 4 these terms are equal and opposite in sign with respect to those appearing in the continuum. This explains why only the Lorentz-invariant form-factors can have non-vanishing coefficients of X. The comparisons with the divergences appearing in the continuum integrals provides therefore another check for our calculation of the Feynman graphs.

The results for the contributions of tables 5 and 6 (which are the only present in the abelian case) are separately gauge-invariant and finite: this means that they escape both checks coming from the vanishing of the sum of the zeroth-order terms in the external momenta and of the comparison with the divergent part of the corresponding integrals in the continuum (indeed these contributions are not present in the continuum and are purely lattice artifacts).

By collecting all the terms we get that the J-part of the expectation value

$$\left\langle \left(\sum_{\rho} F_{\mu\rho} F_{\nu\rho} \right) (-p-q) A_{\alpha}^{a}(p) A_{\beta}^{b}(q) \right\rangle^{\text{irr.}}$$

is given by

$$\begin{aligned}
N\delta^{ab} \Bigg\{ & \left(-\frac{X}{32} + \frac{5Z_0}{96} + \frac{Z_1}{48} \right) (L_2 - L_1) \\
& + \left(\frac{1}{128} - \frac{1}{32\pi^2} - \frac{X}{32} + \frac{5Z_0}{96} - \frac{Z_1}{32} \right) (L_6 - L_4) \\
& + \left(\frac{1}{6} - \frac{5}{24\pi^2} - \frac{11X}{48} + \frac{Z_0}{16} - \frac{7Z_1}{12} \right) (L_9 - L_8) + \left[\left(\frac{11}{192} - \frac{5X}{48} - \frac{59Z_0}{144} - \frac{Z_1}{16} \right) \right. \\
& \left. + (\xi - 1) \left(\frac{X}{32} - \frac{Z_0}{48} \right) \right] (L_{12} - L_{10} - L_{13}) + \left(\frac{1}{64} - \frac{5Z_1}{48} \right) (I_8 - I_3) \\
& + \left(\frac{5}{48} - \frac{5}{24\pi^2} - \frac{2Z_0}{3} - \frac{Z_1}{3} \right) (I_{11} + I_{12} - I_{16}) \Bigg\} \\
& + \frac{2N^2 - 3}{24N} c\delta^{ab} \left[\left(2 - \frac{2}{\pi^2} - 8Z_0 - 6Z_1 \right) (L_9 - L_8) \right. \\
& \left. + \left(-6 + \frac{6}{\pi^2} + 24Z_0 + 14Z_1 \right) (I_{11} + I_{12} - I_{16}) \right] \\
& + \frac{2N^2 - 3}{24N} d\delta^{ab} \left[-2(L_{12} - L_{10} - L_{13}) \right. \\
& \left. + \left(4 - \frac{4}{\pi^2} - 16Z_0 - 4Z_1 \right) (I_{11} + I_{12} - I_{16}) \right].
\end{aligned} \tag{5.9}$$

The J-part of the expectation value

$$\delta_{\mu\nu} \left\langle \left(\sum_{\sigma\rho} F_{\sigma\rho} F_{\sigma\rho} \right) (-p-q) A_{\alpha}^{a}(p) A_{\beta}^{b}(q) \right\rangle^{\text{irr.}}$$

TABLE 1
Results for the insertion of $F_{\mu\alpha}F_{\nu\alpha}$ in the two-gluon correlation function. Contribution of graph (A) of fig. 1

	1	$1/\pi^2$	X	Z_0	Z_1
A_1	−11/4	9/8	0	27/2	47/8
A_2	1/16	−5/8	0	1/2	−3/8
B_1	−51/8	13/2	0	26	31/2
L_1	−1/48	5/48	5/96	−41/288	1/12
L_2	1/16	−5/48	−13/96	53/288	−1/4
L_3	−1/32	5/24	1/12	−11/36	1/8
L_4	0	−1/24	−1/96	29/288	0
L_5	1/32	−5/48	−1/16	7/48	−1/8
L_6	1/128	5/48	1/96	−65/288	0
L_8	−11/96	1/4	11/48	−19/48	11/24
L_9	3/16	−7/24	−3/8	1/2	−3/4
L_{10}	7/192	1/48	−5/48	1/16	−5/24
L_{11}	1/128	−1/48	−1/96	5/288	−1/32
L_{12}	3/64	1/96	7/96	−95/288	−1/8
L_{13}	−1/64	−1/16	−1/24	11/72	0
I_1	17/48	−67/192	0	−227/144	−149/192
I_2	35/48	−7/12	0	−61/18	−37/24
I_3	143/192	−49/48	0	−107/36	−5/3
I_4	−23/192	37/48	0	23/36	−25/48
I_5	−1/64	1/64	0	1/16	3/64
I_6	−1/24	−1/96	0	13/72	13/96
I_7	−55/192	1/8	0	10/9	23/24
I_8	−17/64	5/16	0	11/12	7/8
I_9	5/384	−13/96	0	1/18	−5/96
I_{10}	59/192	−11/32	0	−91/72	−79/96
I_{11}	−37/48	−5/24	0	35/18	13/3
I_{12}	−17/24	−1/12	0	16/9	49/12
I_{13}	91/48	−7/24	0	−101/18	−221/24
I_{14}	−13/128	31/96	0	1/8	13/32
I_{15}	−5/32	3/32	0	13/24	19/32
I_{16}	7/16	−1/48	0	−19/12	−7/4
I_{17}	−13/384	−9/32	0	25/72	13/96

TABLE 2
Results for the insertion of $F_{\mu\alpha}F_{\nu\alpha}$ in the two-gluon correction function. Contribution of graph (B) of fig. 1

	1	$1/\pi^2$	X	Z_0	Z_1
A_1	15/4	−11/4	0	−17	−33/4
A_2	−1/8	5/4	0	−1	3/4
B_1	31/4	−8	0	−32	−18
L_1	7/192	−5/48	−1/48	1/144	−7/48
L_2	−19/384	11/96	1/32	1/288	19/96
L_3	1/48	−1/8	−5/48	5/16	−1/12
L_4	11/256	13/192	1/32	−37/96	−7/64
L_5	−3/128	1/12	5/96	−37/288	3/32
L_6	−1/32	−19/96	−1/48	71/144	1/32
L_{10}	1/64	1/16	3/16	−47/48	5/16
L_{11}	−1/256	1/64	0	0	1/64
L_{12}	−15/128	−1/32	−3/16	61/48	5/32
L_{13}	1/32	1/8	3/16	−43/48	1/8
I_1	−173/384	1/3	0	149/72	23/24
I_2	−187/192	19/32	0	341/72	185/96
I_3	−95/96	37/32	0	293/72	209/96
I_4	83/96	−11/12	0	−31/9	−25/12
I_5	−1/256	1/192	0	0	1/64
I_6	7/384	3/32	0	−13/72	−1/24
I_7	41/384	1/16	0	−59/72	−1/48
I_8	3/64	−5/32	0	−7/24	3/32
I_9	−1/256	−1/192	0	0	1/64
I_{10}	−9/64	1/16	0	1/2	9/16
I_{11}	9/8	−3/8	0	−5/2	−51/8
I_{12}	35/32	−1/2	0	−8/3	−6
I_{13}	−15/8	1	0	4	11
I_{14}	25/256	−21/64	0	−1/8	−25/64
I_{15}	19/128	−3/32	0	−1/2	−19/32
I_{16}	−43/64	5/16	0	9/4	43/16
I_{17}	27/256	9/64	0	−13/24	−27/64

TABLE 3

Results for the insertion of $F_{\mu\alpha}F_{\nu\alpha}$ in the two-gluon correlation function. Contribution of graph (C) of fig. 1 proportional to $N\delta^{ab}$

	1	$1/\pi^2$	X	Z_0	Z_1
A_1	13/24	1/3	0	−11/3	−5/12
A_2	−1/16	−1/8	0	1/2	1/8
B_1	47/24	−25/12	0	−25/3	−49/12
L_1	−1/64	0	0	1/12	1/24
L_2	−5/384	−1/96	7/96	−13/96	7/96
L_3	1/96	−1/12	1/48	−1/144	−1/24
L_4	−1/256	1/192	1/96	−5/288	1/64
L_5	−1/128	1/48	1/96	−5/288	1/32
L_6	0	0	−1/48	5/144	0
L_8	−5/96	−1/24	0	1/3	1/8
L_9	−1/48	1/12	7/48	−7/16	1/6
L_{10}	1/96	−1/12	1/48	−1/144	−1/24
L_{11}	−1/256	1/192	1/96	−5/288	1/64
L_{12}	−1/128	1/48	1/96	−5/288	1/32
L_{13}	0	0	−1/24	5/72	0
I_1	−37/384	17/96	0	29/72	1/6
I_2	−29/192	25/96	0	53/72	13/96
I_3	−5/48	25/96	0	35/72	−1/96
I_4	−23/192	−43/96	0	31/72	85/96
I_5	5/256	−1/48	0	−1/16	−1/16
I_6	7/128	−1/48	0	−1/4	−5/32
I_7	11/128	−1/16	0	−7/24	−5/16
I_8	3/64	−1/32	0	−1/8	−19/96
I_9	−43/768	9/64	0	7/36	31/192
I_{10}	−17/96	7/24	0	29/36	13/48
I_{11}	1/24	1/3	0	−5/18	−1/3
I_{12}	5/48	5/24	0	−4/9	−7/12
I_{13}	−7/12	−25/48	0	85/36	115/48
I_{14}	17/256	−11/192	0	−1/4	−13/64
I_{15}	1/128	0	0	−1/24	0
I_{16}	−1/64	0	0	1/3	−13/48
I_{17}	−43/768	9/64	0	7/36	31/192

TABLE 4

Results for the insertion of $F_{\mu\alpha}F_{\nu\alpha}$ in the two-gluon correlation function. Contribution of graph (D) of fig. 1 proportional to $N\delta^{ab}$

	1	$1/\pi^2$	Z_0	Z_1
A_1	−37/24	31/24	43/6	67/24
A_2	1/8	−1/2	0	−1/2
B_1	−10/3	43/12	43/3	79/12
L_4	−3/64	0	1/4	1/8
L_6	1/32	1/16	−1/4	−1/16
L_{10}	−23/192	0	4/3	0
L_{12}	13/96	0	−4/3	−1/8
L_{13}	−7/96	−1/16	13/12	−1/16
I_1	37/192	−31/192	−43/48	−67/192
I_2	19/48	−13/48	−25/12	−25/48
I_3	1/3	−19/48	−19/12	−19/48
I_4	−5/8	19/32	19/8	55/32
I_6	−1/32	−1/16	1/4	1/16
I_7	3/32	−1/8	0	−5/8
I_8	3/16	−1/8	−1/2	−7/8
I_9	3/64	0	−1/4	−1/8
I_{10}	1/96	−1/96	−1/24	−1/96
I_{11}	−7/24	1/24	1/6	49/24
I_{12}	−37/96	1/6	2/3	13/6
I_{13}	9/16	−3/16	−3/4	−67/16
I_{14}	−1/16	1/16	1/4	3/16
I_{16}	7/48	−1/12	−1/3	−1/3
I_{17}	−1/64	0	0	1/8

TABLE 5
Results for the insertion of $F_{\mu\alpha}F_{\nu\alpha}$ in the two-gluon correlation function. Contribution of graph (C) of fig. 1 proportional to $(2N^2-3)c\delta^{ab}/24N$

	1	$1/\pi^2$	Z_0	Z_1
L_8	−2	2	8	6
L_9	2	−2	−8	−6
I_{11}	−6	6	24	14
I_{12}	−6	6	24	14
I_{16}	6	−6	−24	−14

TABLE 6
Results for the insertion of $F_{\mu\alpha}F_{\nu\alpha}$ in the two-gluon correlation function. Contribution of graph (D) of fig. 1 proportional to $(2N^2-3)d\delta^{ab}/24N$

	1	$1/\pi^2$	Z_0	Z_1
L_{10}	2	0	0	0
L_{12}	−2	0	0	0
L_{13}	2	0	0	0
I_{11}	4	−4	−16	−4
I_{12}	4	−4	−16	−4
I_{16}	−4	4	16	4

is instead

$$N\delta^{ab}\Bigg\{\left(-\frac{1}{8\pi^2}-\frac{3X}{16}+\frac{5Z_0}{16}+\frac{Z_1}{8}\right)(L_2-L_1)$$
$$+\left[\left(\frac{31}{48}-\frac{41}{24\pi^2}-\frac{X}{2}+\frac{5Z_0}{9}-\frac{11Z_1}{4}\right)\right.$$
$$\left.-(\xi-1)\left(\frac{X}{8}-\frac{Z_0}{6}\right)\right](L_9-L_8)\Bigg\}$$
$$+\frac{2N^2-3}{24N}\delta^{ab}\left[(-c+2d)\left(4-\frac{4}{\pi^2}-16Z_0-4Z_1\right)+8d\right](L_9-L_8). \quad (5.10)$$

For the insertion of $T_{\mu\nu}^{\text{gf,tree}}+T_{\mu\nu}^{\text{gh,tree}}$ we show in appendix B that it contributes to the J-integral with

$$N\delta^{ab}\Bigg\{\left[-\frac{1}{192}+\frac{1}{64\pi^2}-\frac{X}{64}+\frac{13Z_0}{192}+(\xi-1)\left(-\frac{1}{192}+\frac{43Z_0}{1152}+\frac{1}{128\pi^2}\right)\right]$$
$$\times(L_2-L_1)$$

$$+\left[\frac{X}{32}-\frac{7Z_0}{96}-(\xi-1)\frac{Z_0}{192}\right](L_6-L_4)$$

$$+\left[\frac{1}{48}-\frac{Z_0}{8}+(\xi-1)\left(\frac{1}{48}-\frac{5Z_0}{36}\right)\right](I_3-I_8)\Bigg\}. \tag{5.11}$$

In order to compute the r.h.s. of the Ward identities we also need the J-contribution of the gluon self-energy $\Pi_{\mu\nu}(p)$, which is explicitly given by

$$N\delta^{ab}\left[-\frac{1}{48}-\frac{1}{48\pi^2}+\left(\frac{5}{12}+\frac{1-\xi}{8}\right)\frac{X}{4}+\left(\frac{7}{72}+\frac{\xi-1}{48}\right)Z_0\right.$$

$$\left.+\frac{2N^2-3}{24N^2}c\right]\left(p^2\delta_{\mu\nu}-p_\mu p_\nu\right). \tag{5.12}$$

Finally we give the tree-level contributions of the operators which can appear in the correction term

$$\begin{aligned}
\left\langle O^{[1]}_{\mu\nu}(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\text{tree}} &= \delta^{ab}(L_9-L_8-L_{10}+L_{12}-L_{13}),\\
\left\langle O^{[2]}_{\mu\nu}(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\text{tree}} &= \delta^{ab}(L_8-L_9),\\
\left\langle O^{[3]}_{\mu\nu}(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\text{tree}} &= \delta^{ab}\left[L_9-L_8-2(I_{11}+I_{12}-I_{16})\right],\\
\left\langle\left(O^{[7]}_{\mu\nu}-O^{[10]}_{\mu\nu}\right)(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\text{tree}} &= \delta^{ab}\left[\tfrac{1}{2}(L_2-L_1)+L_4-L_6\right],\\
\left\langle\left(O^{[8]}_{\mu\nu}-O^{[11]}_{\mu\nu}\right)(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\text{tree}} &= \delta^{ab}\tfrac{1}{2}(L_1-L_2),\\
\left\langle\left(O^{[9]}_{\mu\nu}-O^{[12]}_{\mu\nu}\right)(-p-q)A^a_\alpha(p)A^b_\beta(q)\right\rangle^{\text{tree}} &= \delta^{ab}\left[\tfrac{1}{2}(L_2-L_1)+2(I_8-I_3)\right].
\end{aligned} \tag{5.13}$$

Using these results we get, after some algebra, using (2.25),

$$C_1=Ng^2\left(\frac{7}{192}-\frac{1}{48\pi^2}-\frac{5Z_0}{16}-\frac{Z_1}{16}\right)+\frac{2N^2-3}{24N}g^2(c-2d),$$

$$C_2=-Ng^2\frac{11}{96\pi^2},$$

$$C_3 = Ng^2\left(-\frac{5}{96} + \frac{5}{48\pi^2} + \frac{Z_0}{3} + \frac{Z_1}{6}\right)$$

$$+ \frac{2N^2-3}{24N} g^2\left[c\left(3 - \frac{3}{\pi^2} - 12Z_0 - 7Z_1\right) + d\left(-2 + \frac{2}{\pi^2} + 8Z_0 + 2Z_1\right)\right],$$

$$C_7 = Ng^2\left(-\frac{1}{128} + \frac{1}{32\pi^2} + \frac{Z_0}{64} + \frac{Z_1}{32} + \xi\frac{Z_0}{192}\right),$$

$$C_8 = -Ng^2\frac{3+\xi}{64\pi^2},$$

$$C_9 = Ng^2\left(\frac{1}{128} - \frac{Z_0}{144} - \frac{5Z_1}{96} - \frac{\xi}{96} + \xi\frac{5Z_0}{72}\right). \tag{5.14}$$

For the case of QED the only non-vanishing coefficients are

$$C_1 = e^2(c - 2d),$$

$$C_3 = e^2\left[c\left(3 - \frac{3}{\pi^2} - 12Z_0 - 7Z_1\right) + d\left(-2 + \frac{2}{\pi^2} + 8Z_0 + 2Z_1\right)\right]. \tag{5.15}$$

Numerically we get for U(1) for the case of Wilson's action and the choice $d = 1$

$$C_1 = -e^2,$$

$$C_3 = -0.25997e^2,$$

and for SU(3)

$$C_1 = -0.27076g^2,$$

$$C_3 = 0.03008g^2.$$

These coefficients show that there is a significant correction, of about 30%, for $g^2 \approx 1$, which corresponds to $\beta \approx 6$, which is possible to reach with the present Monte Carlo simulations, for the naive operator $O^{[1]}$. In contrast, the correction due to the non-Lorentz invariant operator $O^{[3]}$ is pretty small, only the 3%. Let's notice that by properly choosing c and d one can set to 0 both C_1 and C_3. In practice c is fixed to 1 by the popular choice of the Wilson action. Still we can use d to obtain a vanishing C_1 or C_3. For instance, fixing in eq. (3.20) $\alpha_2 = 0$ and α_1 equal to -0.1167 for SU(2) or -0.1083 for SU(3) we obtain $C_1 = 0$.

6. Comparison of the lattice with the continuum theory

In this section we shall present a different derivation of the EMT on the lattice which will not make use of the Ward identities of its conservation, but, instead, starts from its definition on the continuum [5,8], which we know gives a conserved quantity, and translate it into a lattice expression.

In general let us consider the following problem: given an operator in the continuum we want to construct the operator on the lattice which gives the same matrix elements. In the continuum theory we will use the dimensional regularization with scale μ and we will renormalize by minimal subtraction. The lattice theory will be renormalized by putting $a = 1/\mu$ *. The two theories differ by a finite renormalization and thus the lattice field A_α has to be renormalized according to

$$A_\alpha = \sqrt{Z_A}\, A_\alpha^{\mathrm{latt}}, \tag{6.1}$$

so that

$$Z_A^{-1} \langle A_\alpha A_\beta \rangle_{a=1/\mu}^{\mathrm{latt}} = \langle A_\alpha A_\beta \rangle_\mu^{\mathrm{cont}}. \tag{6.2}$$

Also the gauge parameter ξ is subject to a finite renormalization as

$$\xi = Z_A \xi^{\mathrm{latt}}. \tag{6.3}$$

At one loop we have

$$\begin{aligned} Z_A = 1 - g^2 N \Bigg[& -\frac{1}{48} - \frac{1}{48\pi^2} + \frac{1}{4}\left(\frac{5}{12} + \frac{1-\xi}{8}\right) Y \\ & + \left(\frac{7}{72} + \frac{\xi - 1}{48}\right) Z_0 - \frac{2N^2 - 3}{24 N^2} c \Bigg], \end{aligned} \tag{6.4}$$

with

$$Y = X(1/\epsilon = 0) = (F_0 - \log 4\pi)/\pi^2. \tag{6.5}$$

Also the coupling constant undergoes a finite renormalization but this is not relevant in our one-loop computation.

* In a previous paper [1] we expressed the operators on the lattice in terms of those of an effective continuum theory equivalent to the lattice one, according to the approach by Symanzik [18]. Thus the continuum operators used in that paper do not have the usual continuum normalization. In practice this means that what we would call in this paper ϕ^2 corresponds to $Z_\phi N[\phi^2]$ of ref. [1].

Given now a generic gauge-invariant operator $\mathscr{O}$ on the continuum we want to find its expression $\mathscr{O}^{\rm latt}$ on the lattice so that

$$\langle \mathscr{O} A_\alpha A_\beta \rangle_\mu^{\rm cont} = \left\langle \mathscr{O}^{\rm latt} A_\alpha^{\rm latt} A_\beta^{\rm latt} \right\rangle_{a=1/\mu}^{\rm latt} = Z_A^{-1} \left\langle \mathscr{O}^{\rm latt} A_\alpha A_\beta \right\rangle_{a=1/\mu}^{\rm latt}. \tag{6.6}$$

The operator $\mathscr{O}^{\rm latt}$ is the sum of a lattice operator $\mathscr{O}^{\rm latt,tree}$ whose formal continuum limit is $\mathscr{O}$ and a correction term which is the sum of all the operators with which $\mathscr{O}$ can mix under renormalization.

Writing at one loop

$$\mathscr{O}^{\rm latt} = \mathscr{O}^{\rm latt,tree} + g^2 \Delta\mathscr{O}, \tag{6.7}$$

and

$$Z_A = 1 + g^2 \Delta Z_A, \tag{6.8}$$

we get from (6.6)

$$g^2 \langle \Delta\mathscr{O} A_\alpha A_\beta \rangle^{\rm latt,tree} = \langle \mathscr{O} A_\alpha A_\beta \rangle^{\rm cont} - \left\langle \mathscr{O}^{\rm latt,tree} A_\alpha A_\beta \right\rangle^{\rm latt} + g^2 \Delta Z_A \langle \mathscr{O} A_\alpha A_\beta \rangle^{\rm cont,tree}. \tag{6.9}$$

In practice, using the method of section 4, we write the expectation value $\langle \mathscr{O}^{\rm latt,tree} A_\alpha A_\beta \rangle^{\rm latt}$ at one loop as $J + K$, where K is the *unrenormalized* continuum contribution. Thus the difference

$$\left\langle \mathscr{O}^{\rm latt,tree} A_\alpha A_\beta \right\rangle^{\rm latt} - \langle \mathscr{O} A_\alpha A_\beta \rangle^{\rm cont} \tag{6.10}$$

is nothing but the J-contribution with $1/\epsilon$ put equal to zero. From formula (6.9) it is easy to identify the operator $\Delta\mathscr{O}$ using the relations (5.13) *.

Let us now give the explicit expression for

$$\begin{aligned} \Delta(F_{\alpha\beta} F_{\alpha\beta} \delta_{\mu\nu}) = Ng^2 \Bigg[&\left(\frac{35}{48} - \frac{13}{8\pi^2} - \frac{11Y}{12} + \frac{Z_0}{6} - \frac{11Z_1}{4} \right) O_{\mu\nu}^{[2]} \\ &+ \left(-\frac{1}{4\pi^2} - \frac{3Y}{8} + \frac{5Z_0}{8} + \frac{Z_1}{4} \right) \left(O_{\mu\nu}^{[8]} - O_{\mu\nu}^{[11]} \right) \Bigg] \\ &+ \frac{2N^2 - 3}{24N} g^2 c \left(-8 + \frac{4}{\pi^2} + 16Z_0 + 4Z_1 \right) O_{\mu\nu}^{[2]} \\ &+ \frac{2N^2 - 3}{24N} g^2 d \left(16 - \frac{8}{\pi^2} - 32Z_0 - 8Z_1 \right) O_{\mu\nu}^{[2]}. \end{aligned}$$

* More precisely we obtain only the gluonic part of $\Delta\mathscr{O}$. The complete determination of $\Delta\mathscr{O}$ would also require the study of ghost correlation functions.

Analogously we can compute

$$
\begin{aligned}
\Delta(F_{\mu\alpha}F_{\nu\alpha}) = Ng^2 \Bigg\{ & \left[\left(-\frac{7}{192} + \frac{1}{48\pi^2} + \frac{5Z_0}{16} + \frac{Z_1}{16} \right) \right] O^{[1]}_{\mu\nu} \\
& + \left(\frac{35}{192} - \frac{7}{24\pi^2} - \frac{11Y}{48} + \frac{Z_0}{24} - \frac{11Z_1}{16} \right) O^{[2]}_{\mu\nu} \\
& - \left(-\frac{5}{96} + \frac{5}{48\pi^2} + \frac{Z_0}{3} + \frac{Z_1}{6} \right) O^{[3]}_{\mu\nu} \\
& - \left(-\frac{1}{128} + \frac{1}{32\pi^2} + \frac{Y}{32} - \frac{5Z_0}{96} + \frac{Z_1}{32} \right) \left(O^{[7]}_{\mu\nu} - O^{[10]}_{\mu\nu} \right) \\
& + \left(-\frac{1}{32\pi^2} - \frac{3Y}{32} + \frac{5Z_0}{32} + \frac{Z_1}{16} \right) \left(O^{[8]}_{\mu\nu} - O^{[11]}_{\mu\nu} \right) \\
& - \left(\frac{1}{128} - \frac{5Z_1}{96} \right) \left(O^{[9]}_{\mu\nu} - O^{[12]}_{\mu\nu} \right) \Bigg\} \\
& - \frac{2N^2 - 3}{24N} g^2 c \left[O^{[1]}_{\mu\nu} - \left(-2 + \frac{1}{\pi^2} + 4Z_0 + Z_1 \right) O^{[2]}_{\mu\nu} \right. \\
& \left. + \left(3 - \frac{3}{\pi^2} - 12Z_0 - 7Z_1 \right) O^{[3]}_{\mu\nu} \right] \\
& - \frac{2N^2 - 3}{24N} g^2 d \left[-2O^{[1]}_{\mu\nu} - \left(4 - \frac{2}{\pi^2} - 8Z_0 - 2Z_1 \right) O^{[2]}_{\mu\nu} \right. \\
& \left. + \left(-2 + \frac{2}{\pi^2} + 8Z_0 + 2Z_1 \right) O^{[3]}_{\mu\nu} \right].
\end{aligned}
\tag{6.11}
$$

Summarizing we get

$$
\begin{aligned}
\Delta T^{\mathrm{gi}}_{\mu\nu} = Ng^2 \Bigg\{ & \left[\left(-\frac{7}{192} + \frac{1}{48\pi^2} + \frac{5Z_0}{16} + \frac{Z_1}{16} \right) \right] O^{[1]}_{\mu\nu} + \frac{11}{96\pi^2} O^{[2]}_{\mu\nu} \\
& - \left(-\frac{5}{96} + \frac{5}{48\pi^2} + \frac{Z_0}{3} + \frac{Z_1}{6} \right) O^{[3]}_{\mu\nu} \\
& - \left(-\frac{1}{128} + \frac{1}{32\pi^2} + \frac{Y}{32} - \frac{5Z_0}{96} + \frac{Z_1}{32} \right) \left(O^{[7]}_{\mu\nu} - O^{[10]}_{\mu\nu} \right) + \frac{1}{32\pi^2} \left(O^{[8]}_{\mu\nu} - O^{[11]}_{\mu\nu} \right)
\end{aligned}
$$

$$- \left(\frac{1}{128} - \frac{5Z_1}{96} \right) \left(O_{\mu\nu}^{[9]} - O_{\mu\nu}^{[12]} \right) \Bigg\}$$

$$- \frac{2N^2-3}{24N} g^2 c \left[O_{\mu\nu}^{[1]} + \left(3 - \frac{3}{\pi^2} - 12Z_0 - 7Z_1 \right) O_{\mu\nu}^{[3]} \right]$$

$$- \frac{2N^2-3}{24N} g^2 d \left[-2O_{\mu\nu}^{[1]} + \left(-2 + \frac{2}{\pi^2} + 8Z_0 + 2Z_1 \right) O_{\mu\nu}^{[3]} \right]. \tag{6.12}$$

We want also to express the continuum operator $\mathscr{T} = T_{\mu\nu}^{\rm gf} + T_{\mu\nu}^{\rm gh}$ in terms of lattice operators. In this case we must take into account the renormalization of ξ and thus we get at one loop

$$g^2 \langle \Delta \mathscr{T} A_\alpha A_\beta \rangle^{\rm latt,tree} = \langle \mathscr{T} A_\alpha A_\beta \rangle^{\rm cont} - \left\langle \mathscr{T}^{\rm latt,tree} A_\alpha A_\beta \right\rangle^{\rm latt}. \tag{6.13}$$

Explicitly we obtain

$$\Delta\left(T_{\mu\nu}^{\rm gf} + T_{\mu\nu}^{\rm gh}\right) = Ng^2 \Bigg\{ \left[\frac{Y}{32} - \frac{7Z_0}{96} - (\xi - 1)\frac{Z_0}{192} \right] \left(O_{\mu\nu}^{[7]} - O_{\mu\nu}^{[10]} \right)$$

$$+ \frac{1+\xi}{64\pi^2} \left(O_{\mu\nu}^{[8]} - O_{\mu\nu}^{[11]} \right)$$

$$+ \left[\frac{1}{96} - \frac{Z_0}{16} + (\xi - 1)\left(\frac{1}{96} - \frac{5Z_0}{72} \right) \right] \left(O_{\mu\nu}^{[9]} - O_{\mu\nu}^{[12]} \right) \Bigg\}. \tag{6.14}$$

Summing these two contributions we finally get the result which agrees with that given in sect. 5.

Let us notice that the final result is on-shell gauge-invariant, although both $\langle \mathscr{O}^{\rm latt,tree} A_\alpha A_\beta \rangle^{\rm latt} - \langle \mathscr{O} A_\alpha A_\beta \rangle^{\rm cont}$ and Z_A are separately ξ-dependent. This cancellation must of course be expected and it has been verified for a generic operator $\mathscr{O}$ in appendix A. Let us also mention that $T^{\rm gf} + T^{\rm gh}$ does not mix with gauge-invariant operators [4]. Thus, if we are interested only in the coefficients C_1, C_2 and C_3 one can avoid the computation of their correlation functions.

7. The trace anomaly

We started from a traceless $T_{\mu\nu}^{\rm tree}$ on the lattice and then corrected it by imposing the validity of Ward identities (2.24). We now show that this procedure gives a tensor whose trace reproduces the correct trace anomaly.

Indeed, as a consequence of (2.24) we get

$$\langle T_{\mu\mu}(0) A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}} = \left[n - 4 + \sum_{i=1}^{n-1} p_i \cdot \frac{\partial}{\partial p_i} \right] \langle A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}} \tag{7.1}$$

On the other hand dimensional analysis tells us that

$$\left[a\frac{\partial}{\partial a} - m\frac{\partial}{\partial m} - \sum_{i=1}^{n-1} p_i \cdot \frac{\partial}{\partial p_i} \right] \langle A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}}$$
$$= (n-4)\langle A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}} \tag{7.2}$$

so that

$$\left[a\frac{\partial}{\partial a} - m\frac{\partial}{\partial m} \right] \langle A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}} = \langle T_{\mu\mu}(0) A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}} \tag{7.3}$$

The change in the cut-off a can be replaced, according to the Callan–Symanzik equation [19], by a change in the parameters m, g, ξ and of the field renormalizations to get that

$$a\frac{\partial}{\partial a}\langle A_{\rho_1}(p_1) \ldots A_{\rho_n}(p_n)\rangle^{\text{irr.}}$$

is given [8] by the insertion at zero momentum of the operator *

$$B = m\gamma_m \bar{\psi}\psi - \frac{\gamma_3}{2\xi} A_\mu \partial_\mu \partial \cdot A - \frac{\beta}{g}\left(2L^\gamma - A_\mu \frac{\delta S}{\delta A_\mu} - \frac{1}{\xi} A_\mu \partial_\mu \partial \cdot A \right) - \tfrac{1}{2}\gamma_3 A_\mu \frac{\delta S}{\delta A_\mu}$$
$$= m\gamma_m \bar{\psi}\psi - \frac{2\beta}{g} L^\gamma + \left(\frac{2\beta}{g} - \gamma_3 \right)\left[\frac{1}{2} A_\mu \frac{\delta S}{\delta A_\mu} + \frac{1}{2\xi} A_\mu \partial_\mu \partial \cdot A \right], \tag{7.4}$$

where

$$\beta(g) = -a\frac{\partial g}{\partial a}, \tag{7.5}$$

$$\gamma_3(g) = a\frac{\partial Z_3}{\partial a}, \tag{7.6}$$

* Clearly this expression would contain also terms proportional to the equation of motion of the ghosts which we neglect as we consider only Green functions with external gluons.

$$\gamma_m(g) = a\frac{\partial Z_m}{\partial a}. \tag{7.7}$$

The $m\partial/\partial m$ term in eq. (7.3) gives the explicit breaking of dilatation invariance. Operator B, which naively should vanish, is the anomaly. At one-loop, in absence of fermions, we have

$$\beta = -\frac{11}{48\pi^2}Ng^3,$$

$$\gamma_3 = -\frac{1}{16\pi^2}g^2N\left[\tfrac{10}{3} + (1-\xi)\right], \tag{7.8}$$

so that the anomaly is given by

$$B = \frac{11}{24\pi^2}Ng^2L^\gamma - Ng^2\frac{3+\xi}{16\pi^2}\left[\tfrac{1}{2}A_\mu\frac{\delta S}{\delta A_\mu} + \frac{1}{2\xi}A_\mu\partial_\mu\partial\cdot A\right]. \tag{7.9}$$

It is easy to see that the trace of the corrected tensor is exactly equal to B. This provides another check for the correctness of our computation.

8. Conclusions

We gave a completely analytical computation of the EMT on the lattice for pure non-abelian gauge theories at the one-loop level.

Such a computation appears necessary, in the sense that naive transcriptions of the EMT on the lattice are not conserved even for the lattice spacing going to zero, and thus they do not produce the correct anomaly and their use gives rise to finite errors.

The one-loop corrected EMT is a good starting point, in the sense that the error goes to zero as the lattice spacing (or equivalently the coupling constant) goes to zero, a fact which is not true for the tree-level EMT since the neglected corrections, due to dimensional transmutation, remain finite in the same limit thus giving a finite error.

In addition, our computation shows that the corrections are already reasonably small ($\approx 30\%$) for the values of the coupling constants used in actual Monte Carlo simulations for SU(3). Moreover, one can exploit the freedom in the definition of the EMT at the tree-level to minimize the corrections, e.g. to the off-diagonal elements. As these off-diagonal elements appear in the matrix-elements used in the analysis of deep-inelastic scattering [15,16,20], the possibility of using them is appealing for future applications to this branch of physics.

We gave ample details of the computational methods and of the intermediate results in the hope to promote further exact perturbative computations on the lattice. For this aim in the appendix C we provide the machinery for all one-loop calculations in pure QCD, while in appendices A and B it is shown how the gauge dependent part of the computation is completely independent of the details of the lattice lagrangian and of the operator at the tree level.

We are grateful to G. Curci for advice in the use of MATHEMATICA, the symbolic manipulation program we used in our computations.

Appendix A. Gauge dependence

In this appendix we present the one-loop ξ-dependent contributions to the renormalization constants. In particular we will show that, whatever is the lattice lagrangian and the specific expression of the energy–momentum tensor, the ξ-dependence is limited to terms which vanish on-shell.

First of all we give some general relations implied by gauge invariance among the various vertices and the propagator. Suppose that the lattice lagrangian has the general form (we set in this appendix $g = 1$)

$$
\begin{aligned}
&-\frac{1}{2}\int\frac{\mathrm{d}p}{(2\pi)^4}\Theta_{\mu\nu}(p)A^a_\mu(p)A^a_\nu(-p)+\frac{1}{3!}\int\frac{\mathrm{d}p}{(2\pi)^4}\frac{\mathrm{d}q}{(2\pi)^4}\frac{\mathrm{d}r}{(2\pi)^4}(2\pi)^4\\
&\times\delta(p+q+r)V^{abc}_{\mu\nu\rho}(p,q,r)A^a_\mu(p)A^b_\nu(q)A^c_\rho(r)\\
&+\frac{1}{4!}\int\frac{\mathrm{d}p}{(2\pi)^4}\frac{\mathrm{d}q}{(2\pi)^4}\frac{\mathrm{d}r}{(2\pi)^4}\frac{\mathrm{d}s}{(2\pi)^4}(2\pi)^4\delta(p+q+r+s)W^{abcd}_{\mu\nu\rho\sigma}(p,q,r,s)\\
&\times A^a_\mu(p)A^b_\nu(q)A^c_\rho(r)A^d_\sigma(s),
\end{aligned}
\tag{A.1}
$$

plus terms with higher powers of the A-field. Then gauge invariance implies the following relations:

$$\hat{k}_\mu\Theta_{\mu\nu}(k)=0,$$

$$\hat{k}_\rho V^{abc}_{\mu\nu\rho}(p,q,k)=if^{abc}\left[\Theta_{\mu\nu}(p)\cos(k_\nu/2)-\Theta_{\mu\nu}(q)\cos(k_\mu/2)\right],$$

$$
\begin{aligned}
\hat{k}_\mu W^{abcd}_{\mu\nu\rho\sigma}(k,p,q,r)&=if^{abc}V^{cde}_{\rho\sigma\nu}(q,r,p+k)\cos(k_\nu/2)\\
&\quad+if^{ace}V^{bde}_{\nu\sigma\rho}(p,r,q+k)\cos(k_\rho/2)
\end{aligned}
$$

$$+if^{ade}V^{bce}_{\nu\rho\sigma}(p, q, r+k)\cos(k_\sigma/2)$$

$$+\tfrac{1}{12}f^{abe}f^{cde}\hat{k}_\nu\big(\delta_{\nu\rho}\Theta_{\rho\sigma}(r)-\delta_{\nu\sigma}\Theta_{\rho\sigma}(q)\big)$$

$$+\tfrac{1}{12}f^{ace}f^{bde}\hat{k}_\rho\big(\delta_{\nu\rho}\Theta_{\nu\sigma}(r)-\delta_{\rho\sigma}\Theta_{\nu\sigma}(p)\big)$$

$$+\tfrac{1}{12}f^{ade}f^{bce}\hat{k}_\sigma\big(\delta_{\nu\sigma}\Theta_{\rho\nu}(q)-\delta_{\rho\sigma}\Theta_{\rho\nu}(p)\big). \qquad \text{(A.2)}$$

Analogously let us consider a generic gauge-invariant operator * $\mathscr{O}(k)$ which has the following expansion in powers of A

$$\frac{1}{2}\int\frac{\mathrm{d}p}{(2\pi)^4}\frac{\mathrm{d}q}{(2\pi)^4}(2\pi)^4\delta(k-p-q)\mathscr{O}^{(2)}_{\mu\nu}(k; p, q)A^a_\mu(p)A^a_\nu(q)$$

$$+\frac{1}{3!}\int\frac{\mathrm{d}p}{(2\pi)^4}\frac{\mathrm{d}q}{(2\pi)^4}\frac{\mathrm{d}r}{(2\pi)^4}(2\pi)^4\delta(k-p-q-r)\mathscr{O}^{(3)abc}_{\mu\nu\rho}(k; p, q, r)$$

$$\times A^a_\mu(p)A^b_\nu(q)A^c_\rho(r)+\frac{1}{4!}\int\frac{\mathrm{d}p}{(2\pi)^4}\frac{\mathrm{d}q}{(2\pi)^4}\frac{\mathrm{d}r}{(2\pi)^4}\frac{\mathrm{d}s}{(2\pi)^4}$$

$$\times(2\pi)^4\delta(k-p-q-r-s)\mathscr{O}^{(4)abcd}_{\mu\nu\rho\sigma}(k; p, q, r, s)$$

$$\times A^a_\mu(p)A^b_\nu(q)A^c_\rho(r)A^d_\sigma(s). \qquad \text{(A.3)}$$

plus terms with higher powers of the A-field.

Gauge invariance implies

$$\hat{p}_\mu\mathscr{O}^{(2)}_{\mu\nu}(k; p, q)=0$$

$$i\hat{q}_\tau\mathscr{O}^{(3)abc}_{\rho\sigma\tau}(p; r, s, q)=f^{abc}\big[\mathscr{O}^{(2)}_{\rho\sigma}(p; r, q+s)\cos(q_\sigma/2)$$

$$-\mathscr{O}^{(2)}_{\sigma\rho}(p; s, q+r)\cos(q_\rho/2)\big],$$

$$i\hat{q}_\alpha\mathscr{O}^{(4)abcd}_{\rho\sigma\tau\alpha}(p; r, s, t, q)=f^{cde}\mathscr{O}^{(3)abe}_{\rho\sigma\tau}(p; r, s, t+q)\cos(q_\tau/2)$$

$$+f^{ade}\mathscr{O}^{(3)bce}_{\sigma\tau\rho}(p; s, t, r+q)\cos(q_\rho/2)$$

$$+f^{bde}\mathscr{O}^{(3)cae}_{\tau\rho\sigma}(p; t, r, s+q)\cos(q_\sigma/2)$$

$$-\tfrac{1}{12}if^{abe}f^{cde}\hat{q}_\tau\big(\mathscr{O}^{(2)}_{\rho\sigma}(p; r, p-r)\delta_{\sigma\tau}$$

* $\mathscr{O}$ can carry arbitrary Lorentz quantum numbers and they will be hidden in the notation.

$$
\begin{aligned}
&-\mathscr{O}^{(2)}_{\rho\sigma}(p;\, p-s,\, s)\delta_{\rho\tau}\Big) \\
&-\tfrac{1}{12} i f^{ace} f^{bde} \hat{q}_\sigma\Big(\mathscr{O}^{(2)}_{\rho\tau}(p;\, r,\, p-r)\delta_{\sigma\tau} \\
&-\mathscr{O}^{(2)}_{\rho\tau}(p;\, p-t,\, t)\delta_{\rho\sigma}\Big) \\
&-\tfrac{1}{12} i f^{bce} f^{ade} \hat{q}_\rho\Big(\mathscr{O}^{(2)}_{\sigma\tau}(p;\, s,\, p-s)\delta_{\rho\tau} \\
&-\mathscr{O}^{(2)}_{\sigma\tau}(p;\, p-t,\, t)\delta_{\rho\sigma}\Big).
\end{aligned}
\qquad (A.4)
$$

We need also to know the dependence of the tree-level gluon propagator $D_{\mu\nu}(p)$ on the parameter ξ. Since the inverse propagator is given by

$$D^{-1}_{\mu\nu}(p) = \Theta_{\mu\nu}(p) + \frac{1}{\xi}\hat{p}_\mu \hat{p}_\nu, \qquad (A.5)$$

we have

$$\frac{\partial}{\partial\xi} D_{\mu\nu}(p) = -D_{\mu\alpha}(p)\frac{\partial D^{-1}_{\alpha\beta}}{\partial\xi}(p) D_{\beta\nu}(p) = \frac{1}{\xi^2}\hat{p}_\alpha \hat{p}_\beta D_{\mu\alpha}(p) D_{\nu\beta}(p), \qquad (A.6)$$

and

$$\hat{p}_\mu = \hat{p}_\alpha D^{-1}_{\alpha\beta}(p) D_{\beta\mu}(p) = \frac{1}{\xi}\hat{p}^2 \hat{p}_\alpha D_{\mu\alpha}(p). \qquad (A.7)$$

We get

$$\frac{\partial}{\partial\xi} D_{\mu\nu}(p) = \frac{\hat{p}_\mu \hat{p}_\nu}{(\hat{p}^2)^2}, \qquad (A.8)$$

which implies

$$D_{\mu\nu}(p) = D_{\mu\nu}(p)\big|_{\xi=0} + \xi\frac{\hat{p}_\mu \hat{p}_\nu}{(\hat{p}^2)^2}. \qquad (A.9)$$

Then, starting from

$$\delta_{\mu\nu} = D_{\mu\alpha}(p) D^{-1}_{\alpha\nu}(p), \qquad (A.10)$$

we get, equating equal powers of ξ,

$$\hat{p}_\mu D_{\mu\nu}(p)\big|_{\xi=0} = 0 \qquad (A.11)$$

and

$$\Theta_{\alpha\mu}(p)D_{\alpha\nu}(p)\big|_{\xi=0}=\Delta(p)P_{\mu\nu}(p), \tag{A.12}$$

where $\Delta(p)=1/\hat{p}^2$ and $P_{\mu\nu}(p)=\hat{p}^2\delta_{\mu\nu}-\hat{p}_\mu\hat{p}_\nu$.

The identities (A.2), (A.4) and these relations are all we need to study the ξ-dependence of the renormalization constants. It will turn out that the result is independent of the specific lagrangian considered and it will be valid for the Wilson lagrangian and for Symanzik improved lagrangians as well.

We want first of all study the ξ-dependence of the correlation function $\langle\mathscr{O}(r)A^a_\alpha(p)A^b_\beta(q)\rangle$.

Because of the transversality property of $\mathscr{O}^{(2)}_{\mu\nu}(k;\,p,\,q)$ there are no ξ^3 contributions. For the same reason, terms proportional to ξ^2 come only from graph B of fig. 1. The contribution is immediately seen to be equal to

$$-\tfrac{1}{2}N\delta^{ab}\mathscr{O}^{(2)}_{\alpha\gamma}(p+q;\,p,\,q)\Theta_{\beta\delta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}\cos\left(\frac{l+q}{2}\right)_\gamma$$

$$\times\cos\left(\frac{l+q}{2}\right)_\delta\hat{l}_\gamma\hat{l}_\delta\Delta(l)^2\Delta(l+q)^2+(\alpha p\leftrightarrow\beta q). \tag{A.13}$$

If $\mathscr{O}$ has dimension four then the order-ξ^2 contribution is exactly equal to the continuum one.

Let us now consider the contributions proportional to ξ. Using the relations (A.2), (A.4) we get for the various diagrams

Graph A of fig. 1:

$$\frac{N}{2}\delta^{ab}\Theta_{\alpha\gamma}(p)\Theta_{\beta\delta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}\mathscr{O}^{(2)}_{\rho\sigma}(p+q;\,l+p+q,\,-l)\Delta(l+q)^2\cos\left(\frac{l+q}{2}\right)_\gamma$$

$$\times\cos\left(\frac{l+q}{2}\right)_\delta D_{\sigma\delta}(l)\big|_{\xi=0}D_{\rho\gamma}(p+q+l)\big|_{\xi=0}$$

$$-N\delta^{ab}\Theta_{\alpha\gamma}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}\mathscr{O}^{(2)}_{\rho\beta}(p+q;\,l+p+q,\,-l)\Delta(l+q)^2\cos\left(\frac{l+q}{2}\right)_\gamma$$

$$\times\cos\left(\frac{l+q}{2}\right)_\beta D_{\rho\gamma}(p+q+l)\big|_{\xi=0}$$

$$+\frac{N}{2}\delta^{ab}\int\frac{\mathrm{d}l}{(2\pi)^4}\mathscr{O}^{(2)}_{\alpha\beta}(p+q;\,l+p+q,\,-l)\Delta(l+q)^2\cos\left(\frac{l+q}{2}\right)_\alpha$$

$$\times\cos\left(\frac{l+q}{2}\right)_\beta+(\alpha p\leftrightarrow\beta q). \tag{A.14}$$

Graph B of fig. 1:

$$-N\delta^{ab}\mathscr{O}^{(2)}_{\alpha\gamma}(p+q;\, p,\, q)\Theta_{\beta\delta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l+q)^2\cos\left(\frac{l+q}{2}\right)_\gamma$$

$$\times\cos\left(\frac{l+q}{2}\right)_\delta D_{\gamma\delta}(l)\big|_{\xi=0}+N\delta^{ab}\mathscr{O}^{(2)}_{\alpha\gamma}(p+q;\, p,\, q)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l)\Delta(l+q)^2$$

$$\times\cos\left(\frac{l+q}{2}\right)_\beta\cos\left(\frac{l+q}{2}\right)_\gamma P_{\beta\gamma}(l)$$

$$+N\delta^{ab}\Theta_{\beta\gamma}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}\mathscr{O}^{(2)}_{\delta\alpha}(p+q;\, -l,\, p+q+l)\Delta(l+q)^2$$

$$\times\cos\left(\frac{l+q}{2}\right)_\gamma\cos\left(\frac{l+q}{2}\right)_\alpha D_{\gamma\delta}(l)\Big|_{\xi=0}$$

$$-N\delta^{ab}\int\frac{\mathrm{d}l}{(2\pi)^4}\mathscr{O}^{(2)}_{\alpha\beta}(p+q;\, p+q+l,\, -l)\ \Delta(l+q)^2$$

$$\times\cos\left(\frac{l+q}{2}\right)_\alpha\cos\left(\frac{l+q}{2}\right)_\beta+(\alpha p\leftrightarrow\beta q). \tag{A.15}$$

Graph C of fig. 1 gives no contribution.
Graph D of fig. 1:

$$N\delta^{ab}\int\frac{\mathrm{d}l}{(2\pi)^4}\mathscr{O}^{(2)}_{\alpha\beta}(p+q;\, p-l,\, q+l)\ \Delta(l)^2\cos\frac{l_\alpha}{2}\cos\frac{l_\beta}{2}$$

$$-N\delta^{ab}\mathscr{O}^{(2)}_{\alpha\beta}(p+q;\, p,\, q)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l)^2\left(\cos^2\frac{l_\alpha}{2}+\frac{1}{12}\hat{l}^2_\alpha\right). \tag{A.16}$$

Suppose now that $\mathscr{O}$ has dimension 4. Then collecting everything together, we get for the second-order expansion in p and q

$$N\delta^{ab}\mathscr{O}^{(2)}_{\alpha\gamma}(p+q;\, p,\, q)\left[\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l)^3P_{\beta\gamma}(l)\cos\left(\frac{l}{2}\right)_\gamma\cos\left(\frac{l}{2}\right)_\beta\right.$$

$$\left.-\tfrac{1}{2}\delta_{\beta\gamma}\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l)^2\left(\cos^2\frac{l_\alpha}{2}+\frac{1}{12}\hat{l}^2_\alpha\right)\right]+(\alpha p\leftrightarrow\beta q)$$

$$=N\delta^{ab}\mathscr{O}^{(2)}_{\alpha\beta}(p+q;\, p,\, q)\left[-\frac{1}{16\pi^2\epsilon}+\frac{1}{32\pi^2}(F_0-\log 4\pi)-\tfrac{1}{48}Z_0\right]. \tag{A.17}$$

Let us notice that this result is completely independent of the form of the lagrangian and of the explicit discretization of the operator.

We want now to compute the ξ-dependent part of the gluon self-energy $\Pi_{\mu\nu}(p)$. There is a ξ^2-contribution equal to

$$\frac{N}{2}\delta^{ab}\Theta_{\mu\rho}(p)\Theta_{\nu\sigma}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l+p)^2\Delta(l)^2(\widehat{l+p})_\rho(\widehat{l+p})_\sigma\cos\frac{l_\rho}{2}\cos\frac{l_\sigma}{2},\tag{A.18}$$

which does not give any contribution when expanded to second order in p.

The contribution of order ξ has the form

$$\frac{N}{2}\delta^{ab}\Theta_{\mu\sigma}(p)\Theta_{\nu\rho}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l+p)^2\cos\left(\frac{l+p}{2}\right)_\rho\cos\left(\frac{l+p}{2}\right)_\sigma D_{\rho\sigma}(l)\bigg|_{\xi=0}$$
$$-N\delta^{ab}\Theta_{\mu\sigma}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l+p)^2\Delta(l)P_{\nu\sigma}(p)\cos\left(\frac{l+p}{2}\right)_\nu\cos\left(\frac{l+p}{2}\right)_\sigma$$
$$+\frac{N}{2}\delta^{ab}\Theta_{\mu\nu}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}\Delta(l)^2\left(\cos^2\frac{l_\mu}{2}+\frac{1}{12}\hat{l}_\mu^2\right)+(\mu\leftrightarrow\nu).\tag{A.19}$$

Expanding to second order in p we obtain

$$N\delta^{ab}\Theta_{\mu\nu}(p)\left[\frac{1}{16\pi^2\epsilon}-\frac{1}{32\pi^2}(F_0-\log 4\pi)+\tfrac{1}{48}Z_0\right].\tag{A.20}$$

Let us notice again that this result is completely independent on the explicit form of the lagrangian.

Formulas (A.17) and (A.20) are all we need to prove that the renormalization constants associated to the gauge-invariant operators are ξ-independent. Indeed the correction to the energy–momentum tensor is obtained computing the difference between the l.h.s. and the r.h.s. of the Ward identity (2.25). It is easy to see that the ξ-dependent contribution of $T_{\mu\nu}^{\mathrm{gi}}$ exactly cancels the contribution of the gluon self-energy and thus we remain with the contributions due to $T_{\mu\nu}^{\mathrm{gf}}$ and $T_{\mu\nu}^{\mathrm{gh}}$. But correlation functions of BRS-variations vanish on-shell and thus no ξ-dependent correction term is needed on-shell.

The second-order expansion of $\langle(T_{\mu\nu}^{\mathrm{gf}}+T_{\mu\nu}^{\mathrm{gh}})(-p-q)A_\alpha^a(p)A_\beta^b(q)\rangle$ is given by formula (B.9) in appendix B. The ξ-dependent contribution is given by

$$-N\delta^{ab}\Theta_{\beta\zeta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l;\,-l)\left(P_{\alpha\rho}(l)\hat{l}_\zeta+P_{\alpha\zeta}(l)\hat{l}_\rho\right)\cos\frac{l_\zeta}{2}\cos\frac{l_\rho}{2}\Delta^3(l)$$

$$+ N\delta^{ab}\Theta_{\alpha\gamma}(p)\int\frac{dl}{(2\pi)^4}K_\rho(l;\,-l)\hat{l}_\rho\hat{l}_\beta\hat{l}_\gamma\cos\frac{l_\beta}{2}\cos\frac{l_\gamma}{2}\Delta^4(l)$$

$$+\frac{N}{12}\delta^{ab}\Theta_{\alpha\beta}(q)\int\frac{dl}{(2\pi)^4}K_\rho(l;\,-l)\hat{l}_\rho\hat{l}_\alpha^2\Delta(l) + (\alpha p a \leftrightarrow \beta q b) \quad (\text{A.21})$$

which is completely independent of the particular form of the lagrangian. By using the explicit form of $K(p,\,q)$ we get

$$\left(-\frac{1}{192}+\frac{1}{128\pi^2}+\frac{43}{1152}Z_0\right)(L_2-L_1)-\frac{1}{192}Z_0(L_6-L_4)$$

$$+\left(\frac{1}{48}-\frac{5}{36}Z_0\right)(I_3-I_8). \quad (\text{A.22})$$

Appendix B Contributions from the gauge-dependent part of the EMT

In this appendix we discuss the computation of the correlation function $\langle (T^{\mathrm{gf}}_{\mu\nu}+T^{\mathrm{gh}}_{\mu\nu})(-p-q)A^a_\alpha(p)A^b_\beta(q)\rangle$.

We begin by considering a generic operator Ξ (with arbitrary Lorentz structure which will be hidden in the notation) of the form

$$\Xi(k) = i\int\frac{dp}{(2\pi)^4}\frac{dq}{(2\pi)^4}(2\pi)^4\delta(k-p-q)K_\alpha(p;\,q)A^a_\alpha(p)\bar{c}^a(q). \quad (\text{B.1})$$

We want now to derive a general formula for the insertion at one-loop of $\delta_{\mathrm{BRS}}\Xi$ in the two-gluon correlation function. The Feynman rules for $\delta_{\mathrm{BRS}}\Xi$ are the following:

$$\begin{aligned}
&\delta_{\mathrm{BRS}}\Xi \to A^a_\alpha(p)A^b_\beta(q): && -\delta^{ab}\left[\hat{q}_\beta K_\alpha(p;\,q)+\hat{p}_\alpha K_\beta(q;\,p)\right]/\xi,\\
&\delta_{\mathrm{BRS}}\Xi \to c^a(p)\bar{c}^b(q): && \delta^{ab}\hat{q}_\alpha K_\alpha(q;\,p),\\
&\delta_{\mathrm{BRS}}\Xi \to c^a(p)\bar{c}^b(q)A^c_\alpha(r): && -if^{abc}\cos(q_\alpha/2)K_\alpha(q+r;\,p),\\
&\delta_{\mathrm{BRS}}\Xi \to c^a(p)\bar{c}^b(q)A^c_\alpha(r)A^d_\beta(t): && -(f^{ace}f^{bde}+f^{ade}f^{bce})\delta_{\alpha\beta}\hat{q}_\alpha\\
& && \times K_\alpha(q+r+t;\,p)/12.
\end{aligned}$$

Let us now compute the various graphs reported in fig. B.1. Using the gauge-invariance relations (A.2) and (A.4) but independently on the explicit form of the gauge-invariant part of the lagrangian, we get

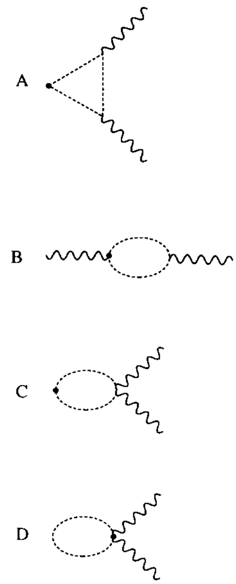

Fig. B.1. Feynman graphs for the insertion of $T^{\mathrm{gh}}_{\mu\nu}$ in the two-gluon correlation function.

Graph A of fig. 1:

$$-if^{bcd}\Theta_{\beta\zeta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)\cos\frac{l_\zeta}{2}$$

$$\times D_{\rho\gamma}(p+q+l)D_{\delta\zeta}(l+q)V^{cda}_{\gamma\delta\alpha}(-p-q-l,\,q+l,\,p)$$

$$+if^{bcd}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)\cos\frac{l_\beta}{2}$$

$$\times D_{\rho\gamma}(p+q+l)V^{cda}_{\gamma\beta\alpha}(-p-q-l,\,q+l,\,p)$$

$$+N\delta^{ab}\Theta_{\alpha\gamma}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)\Delta(l+q)D_{\rho\gamma}(p+q+l)$$

$$\times\left(\widehat{l+q}\right)_\beta\cos\frac{l_\beta}{2}\cos\left(\frac{l+q}{2}\right)_\gamma$$

$$-N\delta^{ab}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)\Delta(l+q)\Delta(l+p+q)$$

$$\times P_{\alpha\rho}(p+q+l)\left(\widehat{l+q}\right)_\beta\cos\frac{l_\beta}{2}\cos\left(\frac{l+q}{2}\right)_\alpha+(\alpha ap\leftrightarrow\beta bq). \qquad \text{(B.2)}$$

Graph C of fig. 1:

$$-if^{bcd}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)$$

$$\times\cos\frac{l_\beta}{2}D_{\rho\gamma}(p+q+l)V^{cda}_{\gamma\beta\alpha}(-p-q-l,\,q+l,\,p)$$

$$+\frac{N}{12}\delta^{ab}\Theta_{\alpha\beta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\hat{l}_\alpha D_{\alpha\rho}(l+p+q)\Delta(l)$$

$$-\frac{N}{12}\delta^{ab}\delta_{\alpha\beta}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\hat{l}_\alpha P_{\alpha\rho}(l+p+q)\Delta(l)\Delta(l+p+q)$$

$$+(\alpha ap\leftrightarrow\beta bq).\tag{B.3}$$

Graph A of fig. B.1:

$$-N\delta^{ab}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)\Delta(l+q)\Delta(l+p+q)$$

$$\times\left(\widehat{l+q}\right)_\beta\left(\widehat{l+p+q}\right)_\alpha\left(\widehat{l+p+q}\right)_\rho\cos\frac{l_\beta}{2}\cos\left(\frac{l+q}{2}\right)_\alpha+(\alpha ap\leftrightarrow\beta bq).\tag{B.4}$$

Graph B of fig. B.1

$$N\delta^{ab}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\alpha(l+p+q;\,-l)\Delta(l)\Delta(l+q)\left(\widehat{l+q}\right)_\beta\cos\frac{l_\beta}{2}\cos\left(\frac{l+q}{2}\right)_\alpha$$

$$+(\alpha ap\leftrightarrow\beta bq).\tag{B.5}$$

Graph C of fig. B.1:

$$-\frac{N}{6}\delta^{ab}\delta_{\alpha\beta}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\,-l)\Delta(l)\Delta(l+p+q)$$

$$\times\hat{l}_\alpha\left(\widehat{l+p+q}\right)_\alpha\left(\widehat{l+p+q}\right)_\rho.\tag{B.6}$$

Graph D of fig. B.1:

$$\frac{N}{6}\delta^{ab}\delta_{\alpha\beta}\int\frac{\mathrm{d}l}{(2\pi)^4}K_\alpha(l+p+q;\,-l)\hat{l}_\alpha\Delta(l).\tag{B.7}$$

Summing up all the contribution we get the result

$$-if^{bcd}\Theta_{\beta\zeta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\ -l)\Delta(l)\cos\frac{l_\zeta}{2}$$

$$\times D_{\rho\gamma}(p+q+l)D_{\delta\zeta}(l+q)V^{cda}_{\gamma\delta\alpha}(-p-q-l,\ q+l,\ p)$$

$$+N\delta^{ab}\Theta_{\alpha\gamma}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\ -l)\Delta(l)\Delta(l+q)$$

$$\times D_{\rho\gamma}(p+q+l)\left(\widehat{l+q}\right)_\beta\cos\frac{l_\beta}{2}\cos\left(\frac{l+q}{2}\right)_\gamma$$

$$+\frac{N}{12}\delta^{ab}\Theta_{\alpha\beta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l+p+q;\ -l)\hat{l}_\alpha D_{\alpha\rho}(l+p+q)\Delta(l)$$

$$+(\alpha ap\leftrightarrow\beta bq). \tag{B.8}$$

Here $\Delta(p)=1/\hat{p}^2$ and $P_{\mu\nu}(p)=\hat{p}^2\delta_{\mu\nu}-\hat{p}_\mu\hat{p}_\nu$.

Expanding to second order in p and q we get

$$-if^{bcd}\Theta_{\beta\zeta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l;\ -l)\Delta(l)\cos\frac{l_\zeta}{2}D_{\rho\gamma}(l)D_{\delta\zeta}(l)V^{cda}_{\gamma\delta\alpha}(-l,\ l,\ 0)$$

$$+N\delta^{ab}\Theta_{\alpha\gamma}(p)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l;\ -l)\Delta(l)^2D_{\rho\gamma}(l)\hat{l}_\beta\cos\frac{l_\beta}{2}\cos\frac{l_\gamma}{2}$$

$$+\frac{N}{12}\delta^{ab}\Theta_{\alpha\beta}(q)\int\frac{\mathrm{d}l}{(2\pi)^4}K_\rho(l;\ -l)\hat{l}_\alpha D_{\alpha\rho}(l)\Delta(l)+(\alpha ap\leftrightarrow\beta bq). \tag{B.9}$$

To go on further we must specify the operator and the explicit form of the lagrangian. For our choice of Ξ (formula (3.22)) K_α is given by

$$K_{\mu\nu,\alpha}(p;\ q)=-\cos(p_\mu/2)\sin q_\nu\ \delta_{\mu\alpha}-\cos(p_\nu/2)\sin q_\mu\ \delta_{\nu\alpha}$$

$$+\cos(p_\alpha/2)\left[\tfrac{1}{2}\sin p_\alpha+\sin q_\alpha\right]\delta_{\mu\nu}. \tag{B.10}$$

Then for the Wilson lagrangian in the Feynman gauge we obtain

$$\left(\underline{\frac{1}{32\pi^2\epsilon}}-\frac{1}{64\pi^2}(F_0-\log 4\pi)+\frac{13}{192}Z_0+\frac{1}{64\pi^2}-\frac{1}{192}\right)(L_2-L_1)$$

$$+\left(-\frac{1}{16\pi^2\epsilon}+\frac{1}{32\pi^2}(F_0-\log 4\pi)-\frac{7}{96}Z_0\right)(L_6-L_4)+\left(\frac{1}{48}-\frac{1}{8}Z_0\right)(I_3-I_8).$$

$$\tag{B.11}$$

Appendix C. Analytic evaluation of lattice integrals

In this appendix we will discuss the computation of the lattice integrals

$$\mathscr{B}(r; n_x, n_y, n_z, n_t) = \int_{-\pi}^{\pi} \left(\frac{\mathrm{d}k}{2\pi}\right)^d \frac{\hat{k}_x^{2n_x} \hat{k}_y^{2n_y} \hat{k}_z^{2n_z} \hat{k}_t^{2n_t}}{\underline{(\hat{k}^2)^r}}, \tag{C.1}$$

with

$$\hat{k}_x = 2 \sin(k_x/2), \tag{C.2}$$

for $n_i \geqslant 0$, $r > 0$, and x, y, z, t all different. In the following when one of the arguments n_i, is zero it will be omitted as argument of $\mathscr{B}$. We will indicate with $\deg \mathscr{B} = n_x + n_y + n_y + n_z - r$ the infrared degree of divergence of $\mathscr{B}$.

These integrals are well defined only if $\deg \mathscr{B} \geqslant -1$, otherwise they are infrared divergent and thus a proper infrared regularization is needed. We will consider here two different types of regularization: one possibility is to dimensionally regularize $\mathscr{B}$, assuming the dimension of the space d to be greater than 4, another one is to replace $\hat{k}^2$ with $\hat{k}^2 + m^2$ in the denominators. We will first consider the dimensional case and then we will extend our considerations to the second case.

Our main result is that each integral $\mathscr{B}(r; n_x, n_y, n_z, n_t)$ can be expressed in terms of only three constants which can be computed numerically.

The first thing we want to show is that each integral $\mathscr{B}(r; n_x, n_y, n_z, n_t)$ with $n_x + n_y + n_z + n_t \neq 0$ can be reduced through purely algebraic manipulations to a set of at most $n_x + n_y + n_z + n_t$ elementary integrals. In order to prove this result let us consider the following integrals

$$\tilde{\mathscr{B}}_\delta(r; n_x, n_y, n_z, n_t) = \int_{-\pi}^{\pi} \left(\frac{\mathrm{d}k}{2\pi}\right)^d \frac{\hat{k}_x^{2n_x} \hat{k}_y^{2n_y} \hat{k}_z^{2n_z} \hat{k}_t^{2n_t}}{(\hat{k}^2)^{r+\delta}}, \tag{C.3}$$

which we will consider for r either positive or negative. Of course

$$\mathscr{B}(r; n_x, n_y, n_z, n_t) = \lim_{\delta \to 0} \tilde{\mathscr{B}}_\delta(r; n_x, n_y, n_z, n_t). \tag{C.4}$$

The integrals $\tilde{\mathscr{B}}_\delta$ satisfy the following recursion relations:

$$\tilde{\mathscr{B}}_\delta(n; 1) = \frac{1}{d} \tilde{\mathscr{B}}_\delta(n-1),$$

$$\tilde{\mathscr{B}}_\delta(n; m, 1) = \frac{1}{d-1} \left[\tilde{\mathscr{B}}_\delta(n-1; m) - \tilde{\mathscr{B}}_\delta(n; m+1) \right],$$

$$\tilde{\mathscr{B}}_\delta(n;\, m,\, r,\, 1) = \frac{1}{d-2}\Big[\tilde{\mathscr{B}}_\delta(n-1;\, m,\, r) - \tilde{\mathscr{B}}_\delta(n;\, m+1,\, r) - \tilde{\mathscr{B}}_\delta(n;\, m,\, r+1)\Big],$$

$$\tilde{\mathscr{B}}_\delta(n;\, m,\, r,\, p,\, 1) = \frac{1}{d-3}\Big[\tilde{\mathscr{B}}_\delta(n-1;\, m,\, r,\, p) - \tilde{\mathscr{B}}_\delta(n;\, m+1,\, r,\, p) - \tilde{\mathscr{B}}_\delta(n;\, m,\, r+1,\, p) - \tilde{\mathscr{B}}_\delta(n;\, m,\, r,\, p+1)\Big], \tag{C.5}$$

which can be obtained by the insertion of the trivial identity

$$\hat{k}^2 = \sum_{q=1}^{d} \hat{k}_q^2 \tag{C.6}$$

and by keeping into account the cases in which the index q equals one of the other indices. Furthermore, when $m > 1$ we can write

$$\frac{\left(\hat{k}_{\underline{w}}^2\right)^m}{\left(\hat{k}^2\right)^{n+\delta}} = 4\frac{\left(\hat{k}_{\underline{w}}^2\right)^{m-1}}{\left(\hat{k}^2\right)^{n+\delta}} + 2\frac{\left(\hat{k}_{\underline{w}}^2\right)^{m-2}}{n+\delta-1}\sin k_w \frac{\partial}{\partial k_w}\frac{1}{\left(\hat{k}^2\right)^{n+\delta-1}}. \tag{C.7}$$

Then, integrating by parts, we obtain the recursion relation

$$\tilde{\mathscr{B}}_\delta(n;\dots,\, m) = \frac{m-1}{(n+\delta-1)}\tilde{\mathscr{B}}_\delta(n-1;\dots,\, m-1) - \frac{4m-6}{n+\delta-1}\tilde{\mathscr{B}}_\delta(n-1;\dots,\, m-2) + 4\tilde{\mathscr{B}}_\delta(n;\dots,\, m-1). \tag{C.8}$$

These relations allow to reduce every integral $\tilde{\mathscr{B}}_\delta(r;\, n_x,\, n_y,\, n_z,\, n_t)$ to a sum of the form

$$\tilde{\mathscr{B}}_\delta(r;\, n_x,\, n_y,\, n_z,\, n_t) = \sum_{p=-\deg \mathscr{B}}^{r-1} \tilde{\mathscr{B}}_\delta(p)\, a_p(\delta). \tag{C.9}$$

For $p \geqslant 1$, $\lim_{\delta\to 0} a_p(\delta)$ is finite while for $p < 1$ they may behave as $1/\delta$ when δ goes to zero.

If $\deg \mathscr{B} \geqslant 0$, terms with $\tilde{\mathscr{B}}_\delta(p)$ with $p \leqslant 0$ appear in the expansion. Our next step is to reduce each $\tilde{\mathscr{B}}_\delta(-p)$, $p \geqslant 1$ to a sum of $\tilde{\mathscr{B}}_\delta(1;\, h)$, $h \geqslant 0$. This can be accomplished through the use of the following recursion relations

$$\tilde{\mathscr{B}}_\delta(-p) = 2(d+2p-2\delta-2)\tilde{\mathscr{B}}_\delta(-p+1) - d(p-\delta-1)\tilde{\mathscr{B}}_\delta(-p+2;\, 2), \tag{C.10}$$

$$\tilde{\mathscr{B}}_\delta(-p;n) = \frac{4n-2}{n}\tilde{\mathscr{B}}_\delta(-p;n-1) + \frac{4}{n}(p-\delta)\left[\tilde{\mathscr{B}}_\delta(-p+1;n) - \frac{1}{4}\tilde{\mathscr{B}}_\delta(-p+1;n+1)\right]. \tag{C.11}$$

The first relation must be used for $p \geqslant 1$, the second one for $p \geqslant 0$ and $n \geqslant 1$.

The first one is obtained writing

$$\int_{-\pi}^{\pi}\left(\frac{\mathrm{d}k}{2\pi}\right)^d (\hat{k}^2)^{p-\delta} = 2d\int_{-\pi}^{\pi}\left(\frac{\mathrm{d}k}{2\pi}\right)^d (\hat{k}^2)^{p-\delta-1}\left(1 - \frac{\partial}{\partial k_x}\sin k_x\right), \tag{C.12}$$

and then integrating by parts, while the second is derived using the identity

$$\left(\hat{k}_x^2\right)^n = 4\left(\hat{k}_x^2\right)^{n-1} - \frac{2}{n-1}\sin k_x \frac{\partial}{\partial k_x}\left(\hat{k}_x^2\right)^{n-1}. \tag{C.13}$$

We thus get

$$\tilde{\mathscr{B}}_\delta(-p) = \alpha(\delta)\tilde{\mathscr{B}}_\delta(0) + \sum_{i=1}^{p+1}\tilde{\mathscr{B}}_\delta(1;i)\beta_i(\delta), \tag{C.14}$$

where $\beta_i(\delta)$ vanish for $\delta \to 0$ as it can be seen from the recursion relations. Thus we obtain eventually

$$\tilde{\mathscr{B}}_\delta(r;n_x,n_y,n_z,n_t) = \sum_{i=0}^{r-1}\theta_i(\delta)\tilde{\mathscr{B}}_\delta(i) + \sum_{i=1}^{\deg\mathscr{B}+1}\zeta_i(\delta)\tilde{\mathscr{B}}_\delta(1;i), \tag{C.15}$$

where all coefficients are finite for $\delta \to 0$, so that we can simply set $\delta = 0$. In this case two terms in the sum simplify, as $\mathscr{B}(0) = 1$ and $\mathscr{B}(1;1) = 1/d$.

Let us now summarize the reduction procedure:

(i) using the recursion relations (C.5) and (C.8) we express all integrals in terms of $\tilde{\mathscr{B}}_\delta(r)$;

(ii) using then the recursion relation (C.10) we reexpress all $\tilde{\mathscr{B}}_\delta(-p)$, $p \geqslant 1$ in terms of $\tilde{\mathscr{B}}_\delta(1;h)$ setting at the end $\delta = 0$.

If $\deg\mathscr{B} \leqslant 0$ the final result depends only on $\mathscr{B}(i)$, $-\deg\mathscr{B} \leqslant i \leqslant r$, while if $\deg\mathscr{B} > 0$ it depends on $\mathscr{B}(i)$, $1 \leqslant i \leqslant r$ and on the $\mathscr{B}(1;p)$ with $1 \leqslant p \leqslant \deg\mathscr{B} + 1$.

We must now deal with the limit $d \to 4$. The integrals $\mathscr{B}(i)$ for $i \geqslant 2$ are infrared divergent and thus we must separate the divergent part from the finite remainder. It follows after a little algebra that

$$\mathscr{B}(r) = \frac{b_{r-2}}{2^r\Gamma(r)}\left(\frac{2}{d-4} - \log 4\pi + \dot{F}_{r-2}\right) + G_{r-2}, \tag{C.16}$$

TABLE C.1
Values of b_r, c_r, G_r, H_r

r	b_r	c_r	G_r	H_r
0	$1/4\pi^2$	0	0	0
1	$1/8\pi^2$	$-1/32\pi^2$	$-7/256\pi^2$	$-1/32\pi^2$
2	$3/32\pi^2$	$-1/32\pi^2$	$-11/1536\pi^2$	$-1/128\pi^2$
3	$13/128\pi^2$	$-55/1536\pi^2$	$-793/589824\pi^2$	$-53/36864\pi^2$
4	$77/512\pi^2$	$-5/96\pi^2$	$-311/1474560\pi^2$	$-331/1474560\pi^2$

where the constants b_r are defined by the asymptotic expansion for large x of $I_0^d(x)$, where $I_0(x)$ is a modified Bessel function [21]

$$I_0^d(x) = \frac{e^{dx}}{(2\pi x)^{d/2-2}} \sum_{i=0}^{\infty} \frac{b_i - (d-4)c_i + O((d-4)^2)}{x^{i+2}} + O(e^{-x}); \quad (C.17)$$

F_p is given by the formula

$$F_p = \frac{1}{b_p} \left\{ \int_0^2 dx\, x^{p+1} \exp(-4x) I_0^4(x) \right.$$
$$\left. + \int_2^{\infty} dx\, x^{p+1} \left[\exp(-4x) I_0^4(x) - \sum_{i=0}^{p} \frac{b_i}{x^{i+2}} \right] \right\}, \quad (C.18)$$

and

$$G_p = \frac{1}{\Gamma(p+2)} \left[\sum_{i=0}^{p-1} \frac{b_i}{2^{i+2}(i-p)} - \frac{c_p}{2^{p+1}} \right]. \quad (C.19)$$

The values of b_k, c_k, G_k for some k are reported in table C.1.

Thus, after this reduction procedure, every integral is expressed in terms of F_p, $p \geqslant 0$ and $\mathscr{B}(1;\, p)$, $p \geqslant 0$. The final step consists in showing that all F_p and $\mathscr{B}(1;\, p)$ can be expressed in terms of only three constants. We will choose

$$Z_0 = \mathscr{B}(1),$$
$$Z_1 = \tfrac{1}{4}\mathscr{B}(1;\, 1, 1), \quad (C.20)$$

and F_0.

To obtain these relations, let us first notice that, for $4m - p + 2 \geqslant 0$ and $p > 0$, the following identities hold

$$\mathscr{B}(p;\, m+1, m, m, m) - \tfrac{1}{4}\mathscr{B}(p-1;\, m, m, m, m) = 0. \quad (C.21)$$

The l.h.s. of these identities can be computed using the method we have discussed. In this way we obtain a set of relations which turn out to be non-trivial and which allow the computation of all $\mathscr{B}(1; r)$, $r \geqslant 3$ and F_p, $p \geqslant 1$ in terms of Z_0, Z_1 and F_0.

Using these relations for $m = 1$ and $1 \leqslant p \leqslant 6$ we get

$$\begin{aligned}
\mathscr{B}(1;3) &= -18 + \frac{12}{\pi^2} + 48 Z_0 + 108 Z_1, \\
\mathscr{B}(1;4) &= \frac{608}{3} - \frac{1216}{9\pi^2} - \frac{1280}{3} Z_0 - \frac{3200}{3} Z_1, \\
\mathscr{B}(1;5) &= -\frac{5663}{3} + \frac{13360}{9\pi^2} + \frac{13760}{3} Z_0 + \frac{29360}{3} Z_1, \\
F_1 &= F_0 - \frac{\pi^2}{8} + \frac{35}{12} + \frac{\pi^2}{2} Z_1, \\
F_2 &= F_0 - \frac{31\pi^2}{144} + \frac{1349}{216} + \frac{\pi^2}{9} Z_0 + \frac{31\pi^2}{36} Z_1, \\
F_3 &= F_0 - \frac{523\pi^2}{1872} + \frac{24257}{2808} + \frac{25\pi^2}{117} Z_0 + \frac{523\pi^2}{468} Z_1.
\end{aligned} \tag{C.22}$$

Let's finally notice that in our previous paper [3] an additional constant was introduced, $Z_2 = \mathscr{B}(1; 1, 1, 1)/8$. This constant can be re-expressed in terms of Z_0 and Z_1 as

$$Z_2 = -\frac{3}{4} + \frac{1}{2\pi^2} + 2Z_0 + \frac{9}{2} Z_1. \tag{C.23}$$

Up to now we have discussed the computation of the integrals $\mathscr{B}$ in dimensional regularization. We want now to discuss the same computation for mass-regularized integrals. We have to consider the integrals

$$\mathscr{B}_m(r; n_x, n_y, n_z, n_t) = \int_{-\pi}^{\pi} \left(\frac{\mathrm{d}k}{2\pi}\right)^4 \frac{\hat{k}_x^{2n_x} \hat{k}_y^{2n_y} \hat{k}_z^{2n_z} \hat{k}_t^{2n_t}}{\left(\hat{k}^2 + m^2\right)^r}. \tag{C.24}$$

TABLE C.2
Numerical values of the three constants Z_0, Z_1 and F_0

Z_0	0.1549333902311
Z_1	0.1077813135399
F_0	4.3692252338748

The recursion relations (C.5), (C.8) and (C.10) are easily extended to this case and thus the same reduction technique can be applied. The only difference is the expression for the divergent integrals $\mathscr{B}_m(p)$, $p \geqslant 2$

$$\frac{1}{\Gamma(p)}\sum_{i=2}^{p-1}\frac{b_{i-2}\Gamma(p-i)}{2^i(m^2)^{p-i}}+\frac{b_{p-2}}{2^p\Gamma(p)}\left(-\log m^2-\gamma_{\mathrm{E}}+F_{p-2}\right)+H_{p-2}, \tag{C.25}$$

where

$$H_p=\frac{1}{\Gamma(p+2)}\sum_{i=0}^{p-1}\frac{b_i}{2^{i+2}(i-p)}. \tag{C.26}$$

We want finally to compare our notations with those used in previous work. Gonzales-Arroyo and Korthals Altes [22] compute lattice integrals in terms of * Z_{0000} and F_{0000} which corresponds respectively to our Z_0 and F_0. In ref. [10] lattice integrals are instead computed in terms of $K = 2Z_0$ and $J = (F_0 - \log(4\pi))/4\pi^2$. Let us finally notice that tables containing the simplest integrals can be found in refs. [11,23].

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* Let us notice that the definition of F_{ijkl}, formula (A.13) of ref. [22] contains a misprint: $(2\pi)^{-4}$ must be substituted by $(16\pi^2)^{-1}$.

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